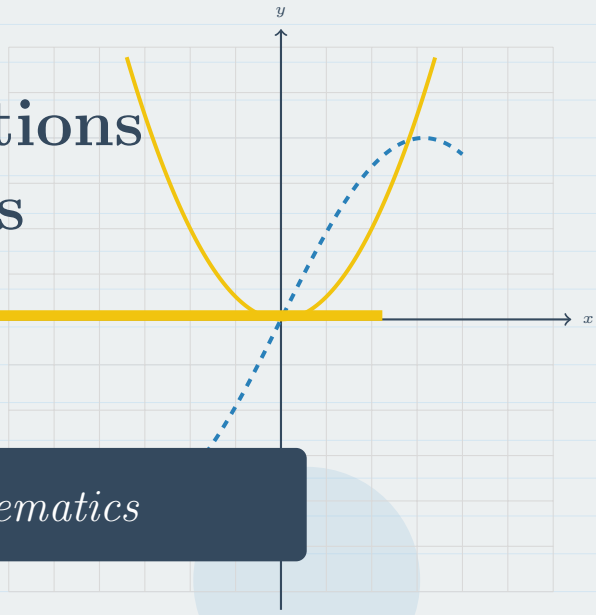
Chapter 3

Graphs, Functions & Calculus



IGCSE O-Level Mathematics

Compiled by:

Dr. Eslam Ahmed

2025 Edition

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1 Sequences

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ZONE MASTERY GUIDE

SEQUENCES & SERIES

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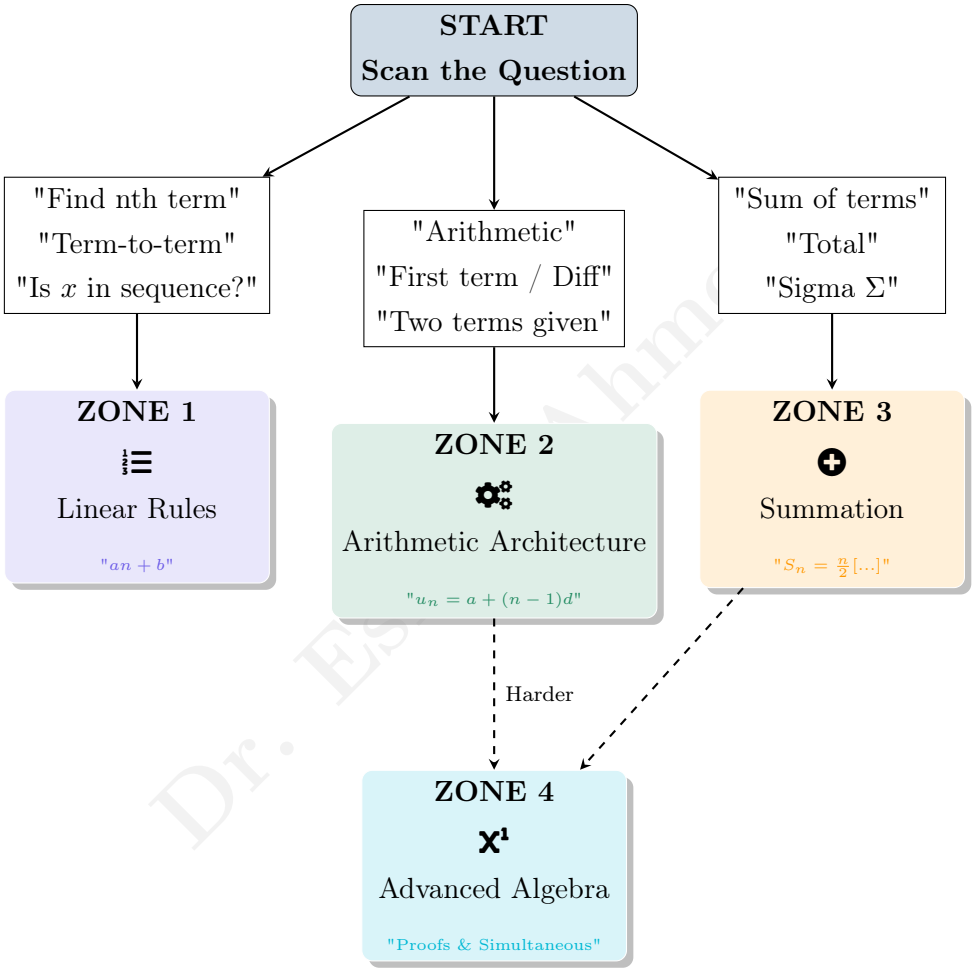
The 5-Second Rule: Match the Trigger → Pick the Zone.



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1 ZONE 1: Linear Rules & The Nth Term

Recognition Zone 1

Zone Name: The Rule Breaker

- **Keywords:** "nth term", "expression in terms of n", "next term", "is 150 in the sequence?".
- **Visual:** A simple list of numbers: $3, 7, 11, 15...$

If you see these → You are in Zone 1.

ZONE 1: The DiNO Method

Strategy: DiNO (Difference, n, Zero-term)

1. **Di (Difference):** Find the gap between terms. This is the coefficient of n.
2. **NO (Zeroth Term):** Go *backwards* one step from the first term. This is the constant.

$$\text{nth term} = (\text{Diff})n + (\text{Zeroth Term})$$

Checking if a number is in the sequence:

- Set your formula equal to the number (e.g., $4n + 1 = 150$).
- Solve for n.
- If n is a **positive integer** (whole number), YES. Otherwise, NO.

Worked Example (Medium Q1)

Problem: Sequence: $3, 9, 15, 21, 27$. (a) Find expression for nth term. (b) Is 150 in the sequence?

→ **Step 1: Find Difference (Di)** $9 - 3 = 6$. The difference is $+6$. Start with $6n$.

→ **Step 2: Find Zeroth Term (NO)** Go back 6 from the first term: $3 - 6 = -3$. Formula: $6n - 3$.

→ **Step 3: Check membership** Set $6n - 3 = 150$.

$$6n = 153$$

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$$n = \frac{153}{6} = 25.5$$

Since 25.5 is **not an integer**, 150 is NOT in the sequence.

Common Traps

- **Trap:** Confusing the term (u_n) with the position (n).
- **Correction:** n is the label (1st, 2nd, 3rd). u_n is the value (3, 9, 15).
- **Trap:** Thinking "negative difference" means the formula is wrong.
- **Correction:** If sequence is $15, 12, 9...$, difference is -3. Formula starts $-3n$.

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2 ZONE 2: Arithmetic Architecture (u_n)

Recognition Zone 2

Zone Name: The Formula Detective

- **Keywords:** "Arithmetic sequence", "First term", "Common difference", "Find a and d".
- **Trigger:** You are given two separated terms (e.g., "6th term is 39, 19th term is 7.8") or consecutive algebraic terms.

If you see these → You are in Zone 2.

ZONE 2: The Simultaneous Protocol

Core Formula: $u_n = a + (n-1)d$

Scenario A: Two non-consecutive terms given

- Write two equations using the formula.
- Subtract the equations to eliminate a and find d.
- Substitute d back to find a.

Scenario B: Three consecutive terms (Algebraic)

- If A, B, C are consecutive, the gap is the same.
- Equation: $B - A = C - B$ (or $2B = A + C$).

Worked Example (Hard Q8 - Finding a & d)

Problem: The 6th term is 39. The 19th term is 7.8. Find the first term.

→ **Step 1: Set up equations** $u_6 = a + 5d = 39 \quad (1)$
$u_{19} = a + 18d = 7.8 \quad (2)$

→ **Step 2: Subtract (2) - (1)**

$$(a + 18d) - (a + 5d) = 7.8 - 39$$

$$13d = -31.2$$

$$d = -2.4$$

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Step 3: Find a Substitute into (1):

$$a + 5(-2.4) = 39$$

$$a - 12 = 39 \implies a = 51$$

Worked Example (Consecutive Algebra)

Problem: $(2x+23), (8x+2), (20x-52)$ are consecutive arithmetic terms. Prove $d = 12$.

Step 1: Equate Differences (2nd) − (1st) = (3rd) − (2nd)

$$(8x + 2) - (2x + 23) = (20x - 52) - (8x + 2)$$

Step 2: Simplify

$$6x - 21 = 12x - 54$$

$$33 = 6x \implies x = 5.5$$

Step 3: Find Terms Term 1: $2(5.5) + 23 = 34$. Term 2: $8(5.5) + 2 = 46$. Difference $d = 46 - 34 = 12$. (Proven)

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3 ZONE 3: Summation Station (S_n)

Recognition Zone 3

Zone Name: The Summer

- **Keywords:** "Sum of first n terms", "Total", "Calculate S_{20}", "Mario saves money".
- **Visual:** $S_n = \dots$ in the formula sheet.

If you see these → You are in Zone 3.

ZONE 3: The Sum Formula

Formula:

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

Or if you know the last term L: $S_n = \frac{n}{2}(a + L)$.

Strategy for "How many terms for Sum > X?"

1. Set $S_n > X$ (or equal to X).
2. Substitute a and d.
3. You will get a **Quadratic Equation** in n.
4. Solve the quadratic (formula or calc). Pick the positive integer n (usually round up).

Worked Example (Medium Q10)

Problem: Sequence $6, 10, 14, 18$. Find sum of first 100 terms.

→ **Step 1: Identify Parameters** $a = 6$. $d = 10 - 6 = 4$. $n = 100$.

→ **Step 2: Apply Formula**

$$S_{100} = \frac{100}{2}[2(6) + (99)(4)]$$

$$S_{100} = 50[12 + 396]$$

$$S_{100} = 50[408] = 20,400$$

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Worked Example (Hard Q4 - Range Sum)

Problem: Sum of terms from 41st to 100th inclusive. Sequence: $a = 1, d = 4$.

→ **Step 1: Concept** Sum(41 to 100) = Sum(first 100) - Sum(first 40).

→ **Step 2: Calculate S_{100} and S_{40}** $S_{100} = \frac{100}{2}[2(1) + 99(4)] = 50[398] = 19900$. $S_{40} = \frac{40}{2}[2(1) + 39(4)] = 20[158] = 3160$.

→ **Step 3: Subtract** $19900 - 3160 = 16740$.

Veteran Tip

Checking Tip: For "Mario saves money" problems (Hard Q6), treat the savings as a sequence. Year 1 = a, Increase = d. "Total" means S_n.

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4 ZONE 4: The Algebra Crunch (Very Hard)

Recognition Zone 4

Zone Name: The Proof Professor

- **Keywords:** "Sum of first 2n terms", "Expression in terms of d", "Prove that...".
- **Trigger:** Mixing S_n and u_n in algebraic equations without numbers.

If you see these → You are in Zone 4.

ZONE 4: Algebraic Substitution

Strategy:

1. Write out the standard formulas for the terms mentioned (S_{2n}, S_n, u_n).
2. Substitute them into the "Worded Equation" given in the question.
3. Expand brackets carefully.
4. Cancel common terms (usually n).
5. Isolate the requested variable.

Worked Example (Very Hard Q1)

Problem: The sum of the first $2n$ terms (S_{2n}) is 4 times the sum of the first n terms (S_n). Find a in terms of d.

→ **Step 1: Write Formulas** $S_n = \frac{n}{2}[2a + (n-1)d]$ $S_{2n} = \frac{2n}{2}[2a + (2n-1)d] = n[2a + (2n-1)d]$

→ **Step 2: Set up Equation** Given $S_{2n} = 4 \times S_n$:

$$n[2a + (2n-1)d] = 4 \times \frac{n}{2}[2a + (n-1)d]$$

→ **Step 3: Simplify** Divide both sides by n:

$$2a + (2n-1)d = 2[2a + (n-1)d]$$

Expand:

$$2a + 2nd - d = 4a + 2nd - 2d$$

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→ **Step 4: Solve** Subtract $2nd$ from both sides (it cancels out!):

$$2a - d = 4a - 2d$$

Rearrange for a:

$$-d + 2d = 4a - 2a$$

$$d = 2a \implies a = \frac{d}{2}$$

Common Traps

- **Trap:** In S_{2n}, replacing n with $2n$ incorrectly.
- **Fix:** S_{2n} means "number of terms" is $2n$. So $\frac{2n}{2}$ outside, and $(2n-1)$ inside the bracket.
- **Trap:** Forgetting that "n" represents a count, so it can usually be divided out of both sides of an algebraic sum equation.

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2 Coordinate Geometry

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ZONE MASTERY GUIDE

COORDINATE GEOMETRY

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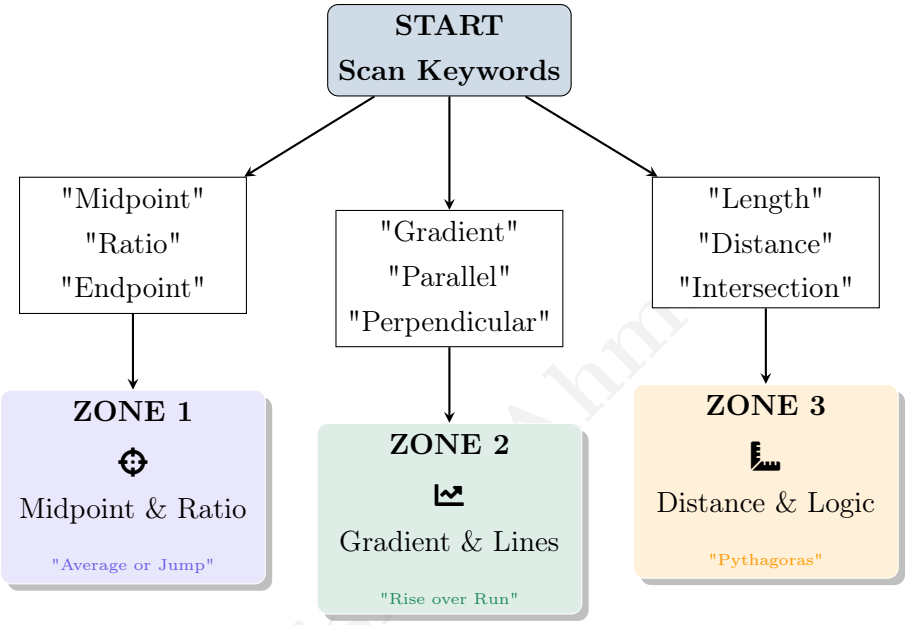
The 5-Second Rule: Match the Trigger → Pick the Zone.



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1 ZONE 1: Midpoints & Endpoints

Recognition Zone 1

Zone Name: The Location Game

- **Keywords:** "Midpoint", "Find coordinates of M", "Endpoint", "Ratio 1 : 2", "Center of the line".
- **Visual:** A line segment with two ends and a point in the middle (or partly along).

If you see these → You are in Zone 1.

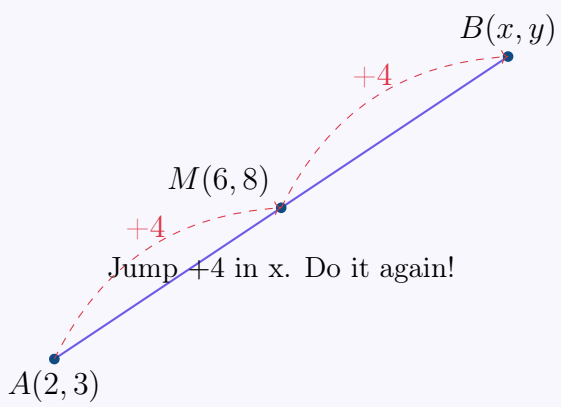
ZONE 1: The Jump Strategy

Strategy A: If finding the Midpoint (Forward) → Step 1: Use the Average Formula The midpoint is simply the average of the coordinates.

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Strategy B: If finding an Endpoint (Reverse) → Step 1: The Jump Method (Recommended) Do NOT use algebra. Use "Jumps".

1. Find the jump from the known endpoint to the midpoint (e.g., +4 in x, −2 in y).
2. Apply the **SAME** jump again from the midpoint to find the other end.



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Worked Example (Reverse Midpoint)

Problem: A is $(2, 3)$. M is the midpoint $(6, 8)$. Find coordinates of B.

Step 1: Find the Jump From A to M:

- x: $2 \to 6$ is a jump of $+\mathbf{4}$.
- y: $3 \to 8$ is a jump of $+\mathbf{5}$.

Step 2: Apply the Jump Again From M to B:

- x: $6 + 4 = \mathbf{10}$
- y: $8 + 5 = \mathbf{13}$

Final Answer

$B(10, 13)$

Common Traps

- **Trap:** Averaging the endpoint and midpoint together.
- **Fix:** Always draw a sketch: $A - - - M - - - B$. If you have M, you must JUMP away from A.

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2 ZONE 2: Gradient & Straight Lines

Recognition Zone 2

Zone Name: The Slope Detectives

- **Keywords:** "Gradient", "Parallel", "Perpendicular", "Equation of line", "y = mx + c".
- **Visual:** Grid lines, straight lines on axes, $y - y_1 = m(x - x_1)$.

If you see these → You are in Zone 2.

ZONE 2: Rise Over Run

Strategy A: Two Points (x_1, y_1) and (x_2, y_2)

$$m = \frac{\text{Change in y}}{\text{Change in x}} = \frac{y_2 - y_1}{x_2 - x_1}$$

Strategy B: Parallel & Perpendicular

- **Parallel:** Gradients are equal ($m_1 = m_2$).
- **Perpendicular:** Gradients are negative reciprocals ($m_1 \times m_2 = -1$).
- *Tip: Flip the fraction and swap the sign.* (e.g., $\frac{2}{3} \to -\frac{3}{2}$)

Worked Example (Perpendicular)

Problem: Line L has equation $y = 2 - 3x$. Find the gradient of a line perpendicular to L.

→ **Step 1: Identify Gradient m_1** Rewrite $y = 2 - 3x$ as $y = -3x + 2$. $m_1 = -3$.

→ **Step 2: Find Perpendicular m_2** Flip -3 (which is $-\frac{3}{1}$) to $\frac{1}{3}$. Swap sign: negative becomes positive.

Final Answer

Gradient $= \frac{1}{3}$

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Veteran Tip

Equation of a Line Trick: Instead of $y = mx + c$, try using:

$$y - y_1 = m(x - x_1)$$

It is often faster if you have a point and a gradient.

Common Traps

- **Trap:** Mixing up the order of subtraction.
- **Fix:** Consistency is key. If you start with Point 2 for y, start with Point 2 for x.

$$\frac{y_2 - y_1}{x_2 - x_1} \quad \text{NOT} \quad \frac{y_2 - y_1}{x_1 - x_2}$$

- **Trap:** Forgetting negative signs. $3 - (-5)$ becomes $3 + 5$.

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3 ZONE 3: Distance, Area & Geometric Logic

Recognition Zone 3

Zone Name: The Surveyor

- **Keywords:** "Length of line", "Distance between", "Area of triangle", "Isosceles", "Perimeter".
- **Visual:** Triangles drawn on grids, diagonal lines, shapes formed by coordinates.

If you see these → You are in Zone 3.

ZONE 3: Pythagoras & The Box Method

Strategy A: Length of a Line It's just Pythagoras!

$$d = \sqrt{(\text{change in x})^2 + (\text{change in y})^2}$$

Tip: You don't need to worry about negative signs inside the square, because $(-3)^2$ *is positive* 9 *anyway.*

Strategy B: Area of Shapes on Grids

- **Easy:** If base and height are horizontal/vertical, use $\frac{1}{2} \times b \times h$.
- **Hard:** If the shape is tilted, draw a rectangle (box) around it, calculate the box area, and subtract the right-angled triangles in the corners.

Worked Example (Reverse Logic)

Problem: Line joining $O(0,0)$ to $P(x,y)$. Gradient is 2. Length is $\sqrt{2645}$. Find P.

→ **Step 1: Use Gradient Information** Gradient $= 2 = \frac{\text{rise}}{\text{run}} = \frac{y}{x}$. Therefore, $y = 2x$.

→ **Step 2: Use Length Information** Length$^2 = x^2 + y^2 = 2645$. Substitute $y = 2x$:

$$x^2 + (2x)^2 = 2645$$

$$x^2 + 4x^2 = 2645 \implies 5x^2 = 2645$$

→ **Step 3: Solve** $x^2 = 529 \implies x = 23$. $y = 2(23) = 46$.

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Final Answer

$P(23, 46)$

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4 ZONE 4: Intersections (The "Very Hard" Questions)

Recognition Zone 4

Zone Name: The Crash Site

- **Keywords:** "Intersect at points A and B", "Solve simultaneously", "Line meets Curve".
- **Visual:** A straight line cutting through a quadratic (U-shape) or circle.

If you see these → You are in Zone 4.

ZONE 4: The Substitution Protocol

The Universal Method:

1. **Isolate:** Make y (or x) the subject of the **Linear** Equation. (e.g., $y = 3x - 1$).
2. **Substitute:** Plug this expression into the **Curve** equation.
3. **Solve:** You will get a quadratic ($ax^2 + bx + c = 0$). Solve for x (factorise or formula).
4. **Return:** Put your x values back into the Linear equation to find y.
5. **Finish:** If asked for midpoint/distance of AB, apply Zone 1 or 3 logic now.

Mega-Problem (Intersection + Midpoint)

Problem: Line $L : x - y = 3$. Curve $C : 3x^2 - y^2 + xy = 9$. Find midpoint of intersection points P and Q.

→ **Step 1: Isolate Linear** From $x - y = 3$, we get $x = y + 3$.

→ **Step 2: Substitute into Curve** Replace every x with $(y + 3)$:

$$3(y + 3)^2 - y^2 + (y + 3)y = 9$$

Expand carefully:

$$3(y^2 + 6y + 9) - y^2 + y^2 + 3y = 9$$

Simplify: $3y^2 + 21y + 18 = 0$. Divide by 3: $y^2 + 7y + 6 = 0$.

→ **Step 3: Solve for y** Factorise: $(y + 6)(y + 1) = 0$. $y_1 = -6, \quad y_2 = -1$.

→ **Step 4: Find x** Using $x = y + 3$: When $y = -6, x = -3 \to P(-3, -6)$. When

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$y = -1, x = 2 \to Q(2, -1)$.

Step 5: Find Midpoint (Zone 1 logic) Average of P and Q: $x : \frac{-3+2}{2} = -0.5$. $y : \frac{-6+(-1)}{2} = -3.5$.

Final Answer

Midpoint $(-0.5, -3.5)$

Veteran Tip

The "Exact Length" Clue: If a question asks for "Exact Length" (like Q4, Page 20), expect your coordinates to be messy or surds. Do NOT convert to decimals. Keep them as $\sqrt{5}$ or fractions.
Length Formula with algebra: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

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3 Functions

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ZONE MASTERY GUIDE

FUNCTIONS: IGCSE MASTER CLASS

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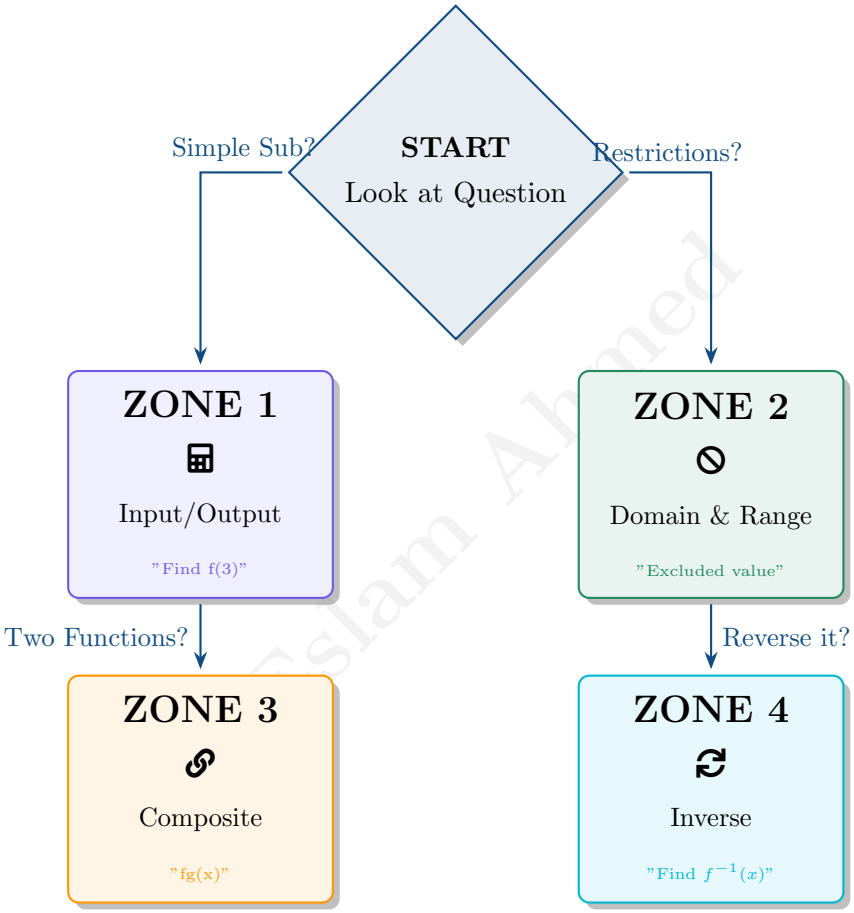
The 5-Second Rule: Match the Trigger → Pick the Zone.



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ZONE 1: The Input/Output Machine

1 ZONE 1: The Input/Output Machine

Functions are machines. You drop a number (x) in the top, the machine performs operations, and spits out an answer (y or $f(x)$).

Recognition: Zone 1

Zone Name: Input/Output Evaluation

Trigger 1: "Find $f(3)$" or "Evaluate $g(-5)$".

Trigger 2: "Find the value of a where $f(a) = 10$".

Trigger 3: Mapping diagrams (arrows connecting numbers).

If you see these → You are in Zone 1.

1.1 Strategy: The Substitution Protocol

→ Step 1: Identify the Input

If the question asks for $f(\boxed{3})$, replace EVERY x in the equation with (3). **→ Step 2: Protect the Input**

Always use brackets, especially for negative numbers.

$$f(x) = x^2 \implies f(-5) = (-5)^2$$

→ Step 3: Solve Backwards (If needed)

If given the **Output** (e.g., $f(x) = 10$), set the equation equal to 10 and solve for x.

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1.2 Worked Examples

Level 1: Basic Substitution (PDF Q1a)

Problem: f and g are functions such that $f(x) = \frac{2}{x^2}$ and $g(x) = 4x^3$. Find $f(-5)$.
Solution:

1. Input is -5. Function is $f(x)$.
2. Substitute with brackets: $f(-5) = \frac{2}{(-5)^2}$
3. Calculate denominator: $(-5)^2 = 25$ (Positive!)
4. Final calculation: $f(-5) = \frac{2}{25}$

Final Answer

$$\frac{2}{25}$$

Level 2: Solving for Input (PDF Q5c)

Problem: $g(x) = \frac{2x}{x+1}$. Given $g(a) = -2$, work out the value of a.
Solution:

1. This is a "Backwards" problem. The output is -2. Input is a.
2. Set up equation: $\frac{2a}{a+1} = -2$
3. Multiply by denominator: $2a = -2(a+1)$
4. Expand: $2a = -2a - 2$
5. Solve: $4a = -2 \implies a = -0.5$

Final Answer

$a = -0.5$

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Common Traps

The Negative Square Trap:

- **Wrong:** $f(x) = x^2 \to f(-3) = -3^2 = -9$
- **Right:** $f(x) = x^2 \to f(-3) = (-3)^2 = 9$

Note: Your calculator thinks -3^2 means $-(3^2)$. Always type $(-3)^2$.

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ZONE 2: The Forbidden Zone (Domain & Range)

2 ZONE 2: Domain & Range

Domain: What goes IN (x). **Range:** What comes OUT (y or $f(x)$).

Recognition: Zone 2

Zone Name: Restricted Values

Trigger 1: "State which value of x must be excluded".

Trigger 2: "State the range".

Trigger 3: Functions with fractions ($\frac{1}{x}$) or roots ($\sqrt{x}$).

If you see these → You are in Zone 2.

2.1 Strategy: Identifying Restrictions

→ Step 1: The Denominator Rule (Domain)

In math, division by zero is illegal. If $f(x) = \frac{a}{bx+c}$, solve $bx + c = 0$. This x is **excluded**.

→ Step 2: The Square Root Rule (Domain)

You cannot root a negative number (in real math). If $f(x) = \sqrt{x-a}$, then $x - a \geq 0$, so domain is $x \geq a$. **→ Step 3: Finding Range (Sketching)**

Do not guess the range. **Sketch the graph**.

- Quadratic (x^2): Has a minimum/maximum vertex. Range is $f(x) \geq$ min.
- Fraction ($\frac{1}{x}$): Cannot equal 0. Range is $f(x) \neq 0$.

2.2 Worked Examples

Level 1: Excluded Domain (PDF Q8b)

Problem: $f(x) = \frac{3}{4+x}$. State which value of x must be excluded from any domain of f.

Solution:

1. Look at the denominator: $4 + x$.
2. Set denominator to zero: $4 + x = 0$.
3. Solve: $x = -4$.
4. If $x = -4$, we divide by zero (Undefined).

Final Answer

$x = -4$

Level 2: Range of a Quadratic (PDF Q10b)

Problem: $f(x) = (x - 4)^2$. State the range of the function f.

Solution:

1. Identify the shape: This is a quadratic (parabola).
2. Logic: Any real number squared is positive or zero.
3. The smallest value $(x - 4)^2$ can be is 0 (when $x = 4$).
4. It can go up to infinity.
5. Therefore, the output must be ≥ 0.

Final Answer

$f(x) \geq 0$ (or $y \geq 0$)

Veteran Tip

Inverse Function Connection: The Domain of $f^{-1}(x)$ is equal to the Range of $f(x)$. The Range of $f^{-1}(x)$ is equal to the Domain of $f(x)$. If asked for the range of an inverse, find the domain of the original!

ZONE 3: Composite Functions (The Chain Reaction)

3 ZONE 3: Composite Functions

Composite functions are like an assembly line. One machine feeds into the next.

Recognition: Zone 3

Zone Name: Chain Reactions

Trigger 1: Notation like $fg(x)$, $gf(x)$, or $h(g(x))$.

Trigger 2: "Find $f^2(x)$" (This means $ff(x)$).

Trigger 3: "Solve $fg(x) = k$".

If you see these → You are in Zone 3.

3.1 Strategy: The Right-to-Left Rule

→ Step 1: Order Matters

ALWAYS work from Right to Left (or Inside to Outside).

$$fg(x) \text{ means } f(g(x)) \implies \text{Do } g \text{ first, then } f.$$

→ Step 2: Numerical Composites (Easy)

If calculating $fg(2)$:

1. Find $g(2)$ (get a number).
2. Take that number and put it into $f(x)$.

→ Step 3: Algebraic Composites (Hard)

If finding $fg(x)$:

1. Take the **entire equation** of $g(x)$.
2. Stuff it into every "x" position inside $f(x)$.
3. Simplify carefully.

3.2 Worked Examples

Level 1: Numerical Composite (PDF Q1b)

Problem: $f(x) = \frac{2}{x^2}$ and $g(x) = 4x^3$. Find $fg(1)$.
Solution:

1. **Right to Left:** Do $g(1)$ first.

$$g(1) = 4(1)^3 = 4(1) = 4$$

2. Now put this result (4) into f.

$$f(4) = \frac{2}{4^2} = \frac{2}{16}$$

3. Simplify.

Final Answer

$\frac{1}{8}$ (or 0.125)

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Level 2: Algebraic Composite (PDF Q2b)

Problem: $f(x) = 3x - 2$ and $g(x) = \frac{10}{x+2}$. Find $gf(x)$.
Solution:

1. **Definition:** $gf(x) = g(f(x))$. This means we put $f(x)$ inside g.
2. Outer function $g(x) = \frac{10}{(\dots)+2}$.
3. Replace the $(\dots)$ with $3x - 2$.

$$g(f(x)) = \frac{10}{(3x-2)+2}$$

4. Simplify the denominator:

$$3x - 2 + 2 = 3x$$

$$g(f(x)) = \frac{10}{3x}$$

Final Answer

$$\frac{10}{3x}$$

Common Traps

The Order Trap: $fg(x)$ is NOT $f(x) \times g(x)$.

- $fg(x)$ means g goes into f.
- $f(x)g(x)$ means multiply them.

Always look for the brackets: $f(g(x))$.

ZONE 4: The Inverse Machine (The Reverse)

4 ZONE 4: Inverse Functions

An inverse function $f^{-1}(x)$ undoes what $f(x)$ did. It maps the Output back to the Input.

Recognition: Zone 4

Zone Name: Reversal

Trigger 1: Notation $f^{-1}(x)$.

Trigger 2: "Express the inverse function...".

Trigger 3: "Find $f^{-1}(5)$" (Tip: You don't always need the equation!).

If you see these → You are in Zone 4.

4.1 Strategy: The Swap Method

→ Step 1: Rename

Replace $f(x)$ with y.

→ Step 2: Swap

Swap every x with y, and every y with x.

→ Step 3: Solve

Rearrange the equation to make the **new** y the subject.

→ Step 4: Rewrite

Replace the final y with $f^{-1}(x)$.

4.2 Worked Examples

Level 1: Linear Inverse (PDF Q2a)

Problem: $f(x) = 3x - 2$. Find $f^{-1}(x)$.

Solution:

1. $y = 3x - 2$
2. Swap: $x = 3y - 2$
3. Add 2: $x + 2 = 3y$
4. Divide by 3: $\frac{x+2}{3} = y$

Final Answer

$$f^{-1}(x) = \frac{x+2}{3}$$

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Level 2: Rational Inverse (PDF Q9b)

Problem: $h(x) = \frac{2x-4}{x}$. Find $h^{-1}(x)$.
Solution:

1. $y = \frac{2x-4}{x}$
2. Swap: $x = \frac{2y-4}{y}$
3. Multiply up: $xy = 2y - 4$
4. **Collect y terms on one side:** $xy - 2y = -4$
5. Factorise y: $y(x-2) = -4$
6. Divide: $y = \frac{-4}{x-2}$
7. Optional Cleanup: Multiply top/bottom by -1: $\frac{4}{2-x}$

Final Answer

$$h^{-1}(x) = \frac{4}{2-x}$$

Veteran Tip

The Mirror Trick: If asked to draw an inverse graph, simply reflect the original line in the diagonal line $y = x$.
The Numerical Trick: If asked to find $f^{-1}(10)$, do NOT find the inverse formula first. Just set $f(x) = 10$ and solve for x.

5 Veteran Challenge: Mixed Zones

Often, IGCSE combines Zones. Here is how to handle a common hybrid.

Zone 3 + Algebra: Solving Composites (PDF Q4d)

Problem: $f(x) = 2x + 5$, $g(x) = x^2 - 25$. Solve $gf(x) = 0$.
Solution:

1. **Construct $gf(x)$:** Put f into g.

$$g(f(x)) = (2x + 5)^2 - 25$$

2. **Set to 0:**

$$(2x + 5)^2 - 25 = 0$$

3. **Solve:** (Don't expand if you don't have to!)

$$(2x + 5)^2 = 25$$

$$2x + 5 = \pm\sqrt{25}$$

$$2x + 5 = \pm 5$$

4. **Case 1:** $2x + 5 = 5 \implies 2x = 0 \implies x = 0$
5. **Case 2:** $2x + 5 = -5 \implies 2x = -10 \implies x = -5$

Final Answer

$x = 0$ or $x = -5$

Conclusion

Functions are procedural.

1. **Input/Output:** Just substitute.
2. **Domain/Range:** Check for zeros in denominators and negative roots.
3. **Composite:** Inside $\to$ Out.
4. **Inverse:** Swap x and y.

Master these four moves, and you master the topic.

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4 Linear Graphs ($y = mx + c$)

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ZONE MASTERY GUIDE

LINEAR GRAPHS $y = mx + c$

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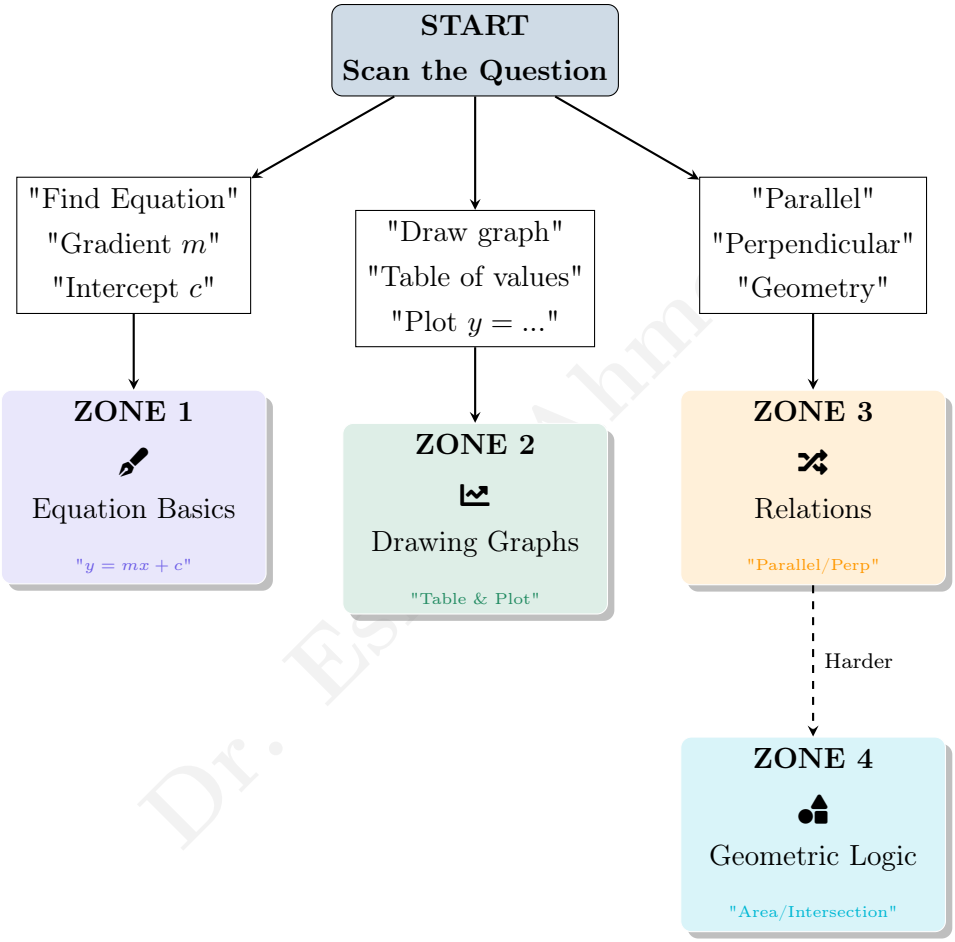
The 5-Second Rule: Match the Trigger → Pick the Zone.



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1 ZONE 1: Equation Basics ($y = mx + c$)

Recognition Zone 1

Zone Name: The Equation Builder

- **Keywords:** "Find the equation", "Gradient", "y-intercept", "Write in form $y = mx + c$".
- **Visual:** A line on a grid without an equation attached.

If you see these → You are in Zone 1.

ZONE 1: The m & c Strategy

Goal: Fill in $y = \mathbf{m}x + \mathbf{c}$.

Step 1: Find m (Gradient)

- Pick two nice integer points on the line.
- Draw a right-angled triangle.
- $m = \frac{\text{change in } y}{\text{change in } x} = \frac{\text{rise}}{\text{run}}$.
- **Check sign:** Uphill is positive $(+)$, Downhill is negative $(-)$.

Step 2: Find c (y-intercept)

- Look where the line crosses the vertical y-axis.
- If you can't see it, substitute a point (x, y) into $y = mx + c$ and solve for c.

Worked Example (Easy Q2)

Problem: Line L passes through $(0, -4)$ and $(2, 0)$. Find equation.

→ **Step 1: Find Gradient (m)** Points: $(0, -4)$ and $(2, 0)$. Rise: $0 - (-4) = 4$. Run: $2 - 0 = 2$. $m = \frac{4}{2} = 2$.

→ **Step 2: Find Intercept (c)** The line crosses the y-axis at -4. So $c = -4$.

→ **Step 3: Assemble** $y = 2x - 4$.

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Common Traps

- **Trap:** Calculating run/rise instead of rise/run.
- **Trap:** Forgetting the negative sign for downhill lines.
- **Trap:** Reading the x-intercept instead of the y-intercept for c.

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2 ZONE 2: Drawing Straight Line Graphs

Recognition Zone 2

Zone Name: The Artist

- **Keywords:** "On the grid, draw the graph", "Complete the table of values".
- **Visual:** An empty grid and an equation like $y = 3x - 2$.

If you see these → You are in Zone 2.

ZONE 2: The Table Method

The Fail-Safe Method:

1. **Table:** Create a small table with x-values (e.g., -2, 0, 2).
2. **Calculate:** Substitute x into the equation to get y.
3. **Plot:** Mark the coordinate pairs (x, y) on the grid.
4. **Join:** Draw a long straight line through them with a ruler.

Worked Example (Easy Q1)

Problem: Draw $y = 3x - 2$ for x from -1 to 3.

→ **Step 1: Table of Values**

x	-1	0	3
y	$3(-1) - 2 = -5$	$3(0) - 2 = -2$	$3(3) - 2 = 7$

→ **Step 2: Plot & Draw** Plot $(-1, -5)$, $(0, -2)$, $(3, 7)$. Connect them.

Veteran Tip

Scale Check: Always look at the axis numbers! 1 square might not equal 1 unit.

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3 ZONE 3: Parallel & Perpendicular Lines

Recognition Zone 3

Zone Name: The Relationship Expert

- **Keywords:** "Parallel", "Perpendicular", "Product of gradients", "Normal".
- **Trigger:** Two lines being compared or one line defined by another.

If you see these → You are in Zone 3.

ZONE 3: The Gradient Rules

Rule 1: Parallel Lines

- **Same Gradient:** $m_1 = m_2$.
- Example: $y = \mathbf{3}x + 2$ is parallel to $y = \mathbf{3}x - 5$.

Rule 2: Perpendicular Lines

- **Negative Reciprocal:** $m_1 \times m_2 = -1$.
- Flip the fraction and swap the sign.
- Example: Gradient $2 \to -\frac{1}{2}$.
- Example: Gradient $-\frac{3}{4} \to \frac{4}{3}$.

Worked Example (Medium Q16)

Problem: Find line perpendicular to $y = 4x - 5$ passing through $(0, 6)$.

→ **Step 1: Identify** m_1 Original line: $y = 4x - 5 \implies m_1 = 4$.

→ **Step 2: Find Perpendicular** m_2 Flip 4 (which is $\frac{4}{1}$) to $\frac{1}{4}$. Change sign to negative. $m_2 = -\frac{1}{4}$.

→ **Step 3: Use Point** $(0, 6)$ Since $x = 0$, the y-intercept $c = 6$. Equation: $y = -\frac{1}{4}x + 6$.

Common Traps

- **Trap:** Using the gradient of the original line for the perpendicular one.
- **Trap:** Not rearranging to $y = mx + c$ first.

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- Example: $2y + x = 5 \implies 2y = -x + 5 \implies y = -\frac{1}{2}x + 2.5$. Gradient is $-\frac{1}{2}$.

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4 ZONE 4: Geometric Logic (Very Hard)

Recognition Zone 4

Zone Name: The Shape Solver

- **Keywords:** "Area of triangle", "Rhombus", "Rectangle", "Coordinates of intersection", "Midpoint".
- **Trigger:** Questions combining lines with shapes or intersections.

If you see these → You are in Zone 4.

ZONE 4: The Intersection & Area Protocol

Strategy A: Intersection Point

- Solve the equations simultaneously.
- Set $y = y$ (e.g., $2x + 1 = -x + 4$).

Strategy B: Area of Triangle formed by lines

- Find the vertices (Intersections with axes and each other).
- Sketch the triangle.
- Use Area $= \frac{1}{2} \times$ base $\times$ height.
- Often base is on x or y axis.

Strategy C: Shapes (Rhombus/Kite)

- Diagonals intersect at $90°$ (Perpendicular).
- Diagonals bisect each other (Midpoint).

Worked Example (Hard Q11 - Rhombus)

Problem: $A(5, 11)$. Diagonal DB is $y = \frac{1}{2}x + 6$. Find equation of diagonal AC.

→ **Step 1: Property Check** Diagonals of a rhombus are perpendicular.

→ **Step 2: Find Gradient of AC** Gradient of DB is $m_1 = \frac{1}{2}$. Perpendicular gradient $m_2 = -2$.

→ **Step 3: Find Equation** Passes through $A(5, 11)$ with $m = -2$. $y - 11 = -2(x - 5)$ $y - 11 = -2x + 10$ $y = -2x + 21$.

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Veteran Tip

Equation Form: If asked for $ax + by + c = 0$, rearrange at the end. Example: $y = \frac{2}{3}x + 4 \xrightarrow{\times 3} 3y = 2x + 12 \implies 2x - 3y + 12 = 0$.

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5 Graphs of Functions

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ZONE MASTERY GUIDE

GRAPHS OF FUNCTIONS

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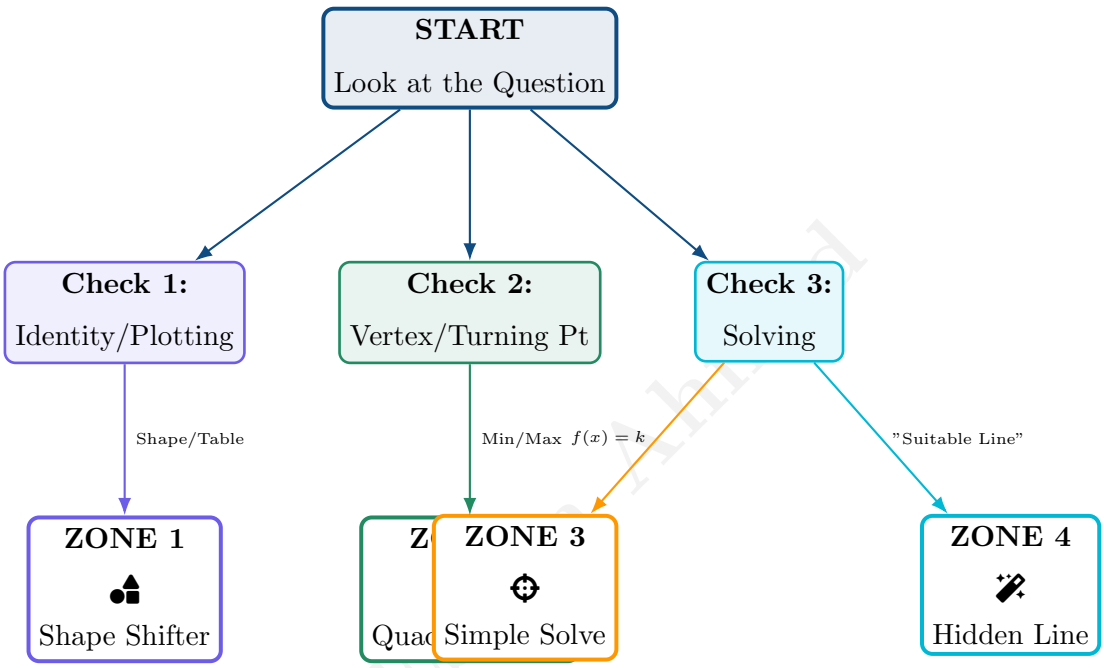
The 5-Second Rule: Match the Trigger → Pick the Zone.



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ZONE 1: The Shape Shifter

Recognition Zone 1

Zone Name: Recognition & Plotting

- **Trigger:** "Complete the table", "Sketch the graph", "Match the equation".
- **Keywords:** Linear, Quadratic, Cubic, Reciprocal, Sin, Cos, Tan.

If you see these → You are in Zone 1.

→ Step 1 Identify the Power

Look at the highest power of x:

- $x^1 \to$ Line.
- $x^2 \to$ Parabola ($\cup$ or $\cap$).
- $x^3 \to$ Snake ($\sim$).
- $\frac{1}{x} \to$ Reciprocal (L-shapes).

→ Step 2 Table Calculation

Use the **Table Mode** on your calculator. **WARNING:** When substituting negative numbers manually, always use brackets! Example: $(-3)^2 = 9$, but $-3^2 = -9$.

Worked Example: Matching Graphs (Hard)

Problem: Match the equations to the graphs. Graph A: Reciprocal (positive), Graph B: Cubic (negative), Graph C: Reciprocal (negative). Equations: (1) $y = \frac{4}{x}$, (2) $y = -0.7x^3$, (3) $y = -\frac{5}{x}$.

Solution:

- $y = \frac{4}{x}$ is Reciprocal (Positive quadrant 1 & 3) $\to$ Graph A.
- $y = -0.7x^3$ is Cubic (Starts top left, ends bottom right) $\to$ Graph B.
- $y = -\frac{5}{x}$ is Reciprocal (Negative quadrant 2 & 4) $\to$ Graph C.

Veteran Tip

For Reciprocal graphs ($y = \frac{a}{x}$), remember the curve NEVER touches the axes. These are asymptotes.

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ZONE 2: Quadratic Anatomy

Recognition Zone 2

Zone Name: Turning Points & Completing Square

- **Trigger:** "Find coordinates of turning point", "Write in form $a(x+p)^2+q$".
- **Keywords:** Vertex, Minimum, Maximum, Symmetry.

If you see these → You are in Zone 2.

Step 1 Complete the Square

To find the Turning Point of $y = x^2 + bx + c$: Write it as $(x + \frac{b}{2})^2 - (\frac{b}{2})^2 + c$.

Step 2 Read the Vertex

From form $y = a(x + p)^2 + q$:

- **x-coordinate:** Change the sign of p ($\to -p$).
- **y-coordinate:** Keep q as is ($\to q$).
- Vertex is at $(-p, q)$.

Worked Example: Finding Turning Point

Problem: Find the turning point of $y = x^2 - 6x + 4$.

Solution:

1. Halve the coefficient of x (which is -6) $\to -3$.
2. Bracket: $(x - 3)^2$.
3. Subtract square: $(x - 3)^2 - (-3)^2 + 4$.
4. Simplify: $(x - 3)^2 - 9 + 4 = (x - 3)^2 - 5$.
5. Vertex: $(3, -5)$.

Common Traps

Students often forget to **subtract** the squared number. They write $(x + 3)^2 + 9$ instead of $(x + 3)^2 - 9$.

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ZONE 3: The Intersection Hunter

Recognition Zone 3

Zone Name: Graphical Solving

- **Trigger:** "Use the graph to solve...", "Find estimates for...".
- **Keywords:** Intersection, Roots, Simultaneous.

If you see these → You are in Zone 3.

→ Step 1 Identify the Target

- If solving $f(x) = k$: Draw horizontal line $y = k$.
- If solving $f(x) = g(x)$: Draw line $y = g(x)$.

→ Step 2 Read x-values

Find where the lines cross. Drop a vertical line to the x-axis. Read the value.

Worked Example: Simple Solve

Problem: Graph of $y = x^2 - 4x - 2$ is drawn. Solve $x^2 - 4x - 2 = 0$.
Solution:

- The equation matches the graph exactly.
- We are solving for $y = 0$ (the x-axis).
- Look at where the curve crosses the x-axis.
- From PDF Q14: Roots are approx $x = -0.4$ and $x = 4.4$.

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ZONE 4: The Phantom Line (Hard)

Recognition Zone 4

Zone Name: Rearrange & Solve

- **Trigger:** "By drawing a suitable straight line...", Equation given $\neq$ Graph drawn.
- **Action:** Subtraction Method.

If you see these → You are in Zone 4.

→ Step 1 Stack the Equations

Write the Equation of the **Graph** on top ($y = \dots$). Write the Equation to **Solve** below it ($0 = \dots$).

→ Step 2 Subtract

Subtract the bottom equation from the top one.

$$(Graph) - (Target) = (Line\ to\ Draw)$$

Worked Example: The Killer Question

Problem: You have the graph of $y = x^3 - 3x + 1$. Solve $x^3 - 3x + 1 = 2x - 1$ by drawing a line.
Strategy: 1. Equation 1 (Graph): $y = x^3 - 3x + 1$
2. Equation 2 (Target): $0 = x^3 - 3x + 1 - (2x - 1)$ rearranged? Wait! The problem says solve $x^3 - 3x + 1 = 2x - 1$. This is simply intersection of Curve and $y = 2x - 1$.
Harder Variation (PDF Q7): Graph: $y = 2x^2 - 4x - 1$. Solve: $x^2 - x - 1 = 0$. 1. Multiply Target by 2 to match x^2 term: $2x^2 - 2x - 2 = 0$. 2. Stack: $y = 2x^2 - 4x - 1$ $0 = 2x^2 - 2x - 2$ 3. Subtract: $y - 0 = (-4x - -2x) + (-1 - -2)$ $y = -2x + 1$ 4. Action: Draw line $y = -2x + 1$. Find intersections.

Veteran Tip

If the x^2 or x^3 coefficients don't match, MULTIPLY the target equation until they do!

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6 Estimating Gradients

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ESTIMATING GRADIENTS OF TANGENTS

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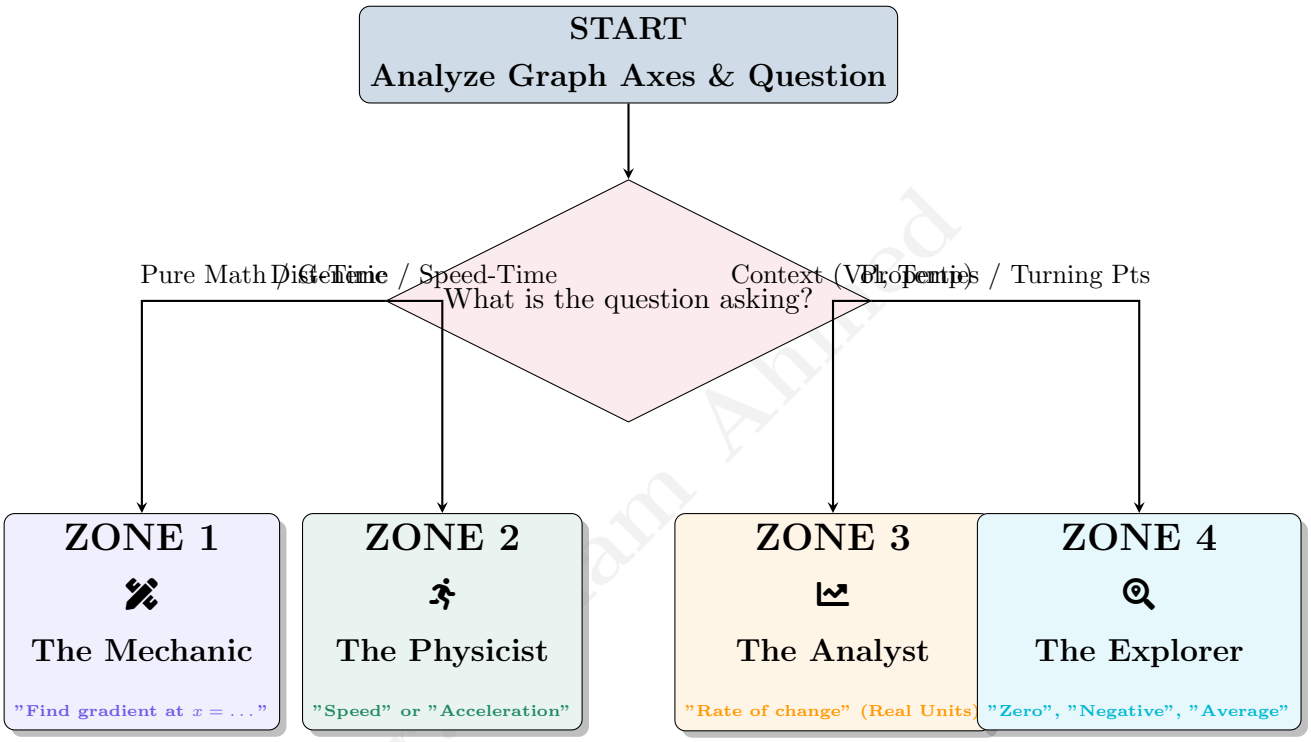
The 5-Second Rule: Match the Trigger → Pick the Zone.



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ZONE 1: The Mechanic (Pure Gradients)

Recognition Zone 1

Zone Name: The Mechanic

- **Trigger:** You see a curve (parabola, cubic, etc.) on $x - y$ axes.
- **Keywords:** "Find an estimate for the gradient", "Draw a tangent", "at the point where $x = \ldots$".
- **Goal:** Physically draw a line and calculate rise/run.

If you see these → You are in Zone 1.

→ Step 1: The Ruler Rule

Place your ruler at the specific x-value point on the curve. Adjust the angle so the ruler touches the curve at **only that single point** locally. The angles between the ruler and the curve on both sides should look equal (like a balanced see-saw).

→ Step 2: Draw Big

Draw a **long** line. Do not draw a tiny dash. Extend it across the grid lines to find easy coordinates.

→ Step 3: The Triangle

Pick two points on your tangent line that are **far apart** and cross grid intersections perfectly if possible.

$$m = \frac{\text{Change in } y}{\text{Change in } x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Veteran Tip

Examiners allow a range of answers (e.g., 2.1 → 2.5). Precision comes from picking points far apart. A tiny triangle leads to huge errors!

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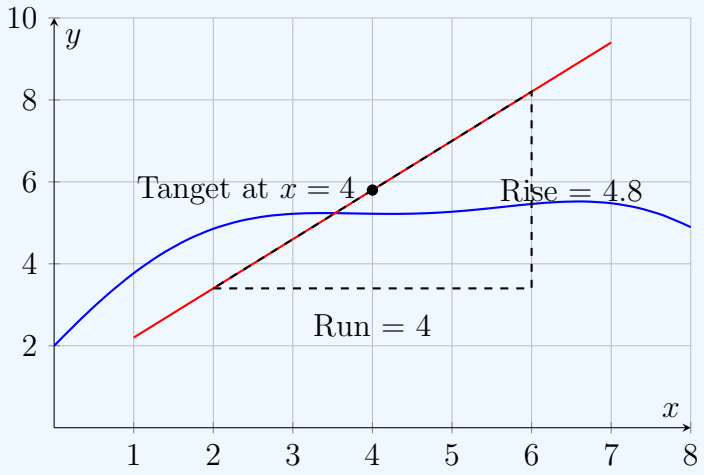
Worked Example: Generic Curve

Problem: The graph shows parts of the curve $y = f(x)$. Calculate an estimate for the gradient of the curve at the point where $x = 4$.



Solution: 1. Draw a tangent at $x = 4$ (Red line). 2. Pick two points on the line: $(2, 3.4)$ and $(6, 8.2)$. 3. Calculate gradient:

$$m = \frac{8.2 - 3.4}{6 - 2} = \frac{4.8}{4} = 1.2$$

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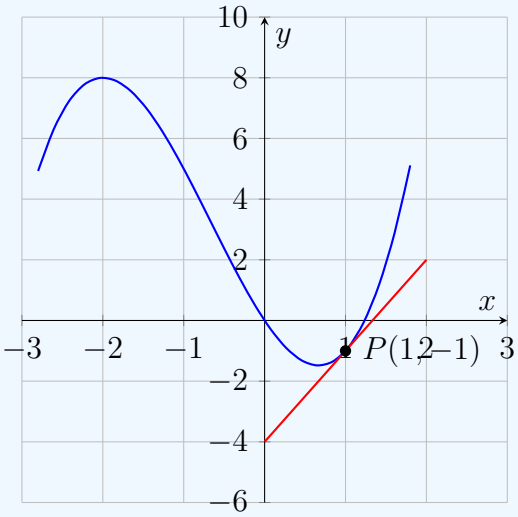
Worked Example: Cubic Curve

Problem: The curve $y = x^3 + 2x^2 - 4x$ is shown. Find an estimate of the gradient when $x = 1$.



Solution: 1. Identify the point $(1, -1)$. 2. Draw the tangent. 3. Pick points $(0, -4)$ and $(2, 2)$.

$$m = \frac{2 - (-4)}{2 - 0} = \frac{6}{2} = 3$$

Final Answer

Estimate of gradient $= 3$

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ZONE 2: The Physicist (Kinematics)

Recognition Zone 2

Zone Name: The Physicist

- **Trigger:** Axes are labeled **Time** (x-axis) and **Distance, Speed, or Velocity** (y-axis).
- **Keywords:** "Estimate the speed", "Estimate the acceleration".
- **Goal:** Translate physics words into "Gradient".

If you see these → You are in Zone 2.

→ Step 1: Translate the Question

- **Distance-Time Graph:** Gradient = **Speed** (m/s).
- **Speed-Time Graph:** Gradient = **Acceleration** (m/s^2).

→ Step 2: Execute Zone 1 Technique

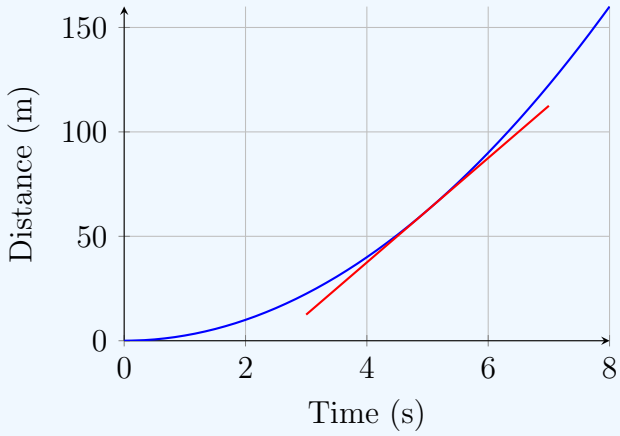
Once you know you need the gradient, follow the Zone 1 method: Draw Tangent → Triangle → Rise/Run.

Common Traps

Curved Graphs vs Straight Lines: If the graph is curved, speed/acceleration is **changing**. You MUST draw a tangent at the specific time requested. Do not just divide the coordinates of the point (y/x) unless the line is straight from the origin!

Worked Example: Distance-Time (Speed)

Problem: The distance-time graph shows a car journey. Estimate the speed at $t = 5$ seconds.

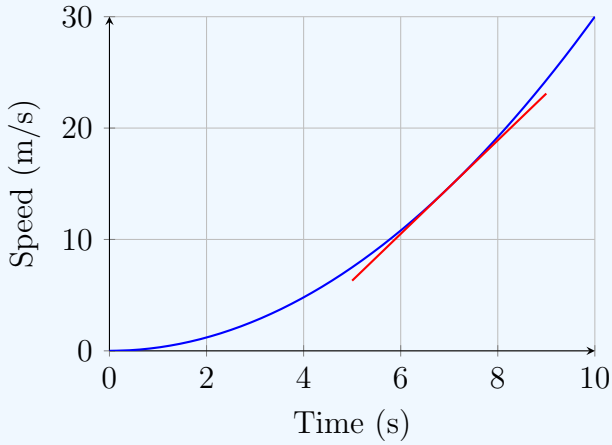


Solution: 1. "Speed" on a Distance-Time graph means **Gradient**. 2. Draw tangent at $t = 5$. 3. Pick points on the red line, e.g., $(4, 37.5)$ and $(6, 87.5)$.

$$\text{Speed} = \frac{87.5 - 37.5}{6 - 4} = \frac{50}{2} = 25 \text{ m/s}$$

Worked Example: Speed-Time (Acceleration)

Problem: The graph shows the speed of a car. Estimate the acceleration at $t = 7$ seconds.



Solution: 1. "Acceleration" on a Speed-Time graph means **Gradient**. 2. Draw tangent at $t = 7$. 3. Pick points from tangent: $(5, 6.3)$ and $(9, 23.1)$.

$$\text{Acceleration} = \frac{23.1 - 6.3}{9 - 5} = \frac{16.8}{4} = 4.2 \text{ m/s}^2$$

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ZONE 3: The Analyst (Real-Life Rates)

Recognition Zone 3

Zone Name: The Analyst

- **Trigger:** The y-axis has real physical units (Volume, Temperature, Depth, Cost) and the question asks for a "Rate of change".
- **Keywords:** "Rate of decrease", "How fast is it filling?", "Instantaneous rate".
- **Goal:** Gradient = Rate. (Check the units!)

If you see these → You are in Zone 3.

→ Step 1: Units Check

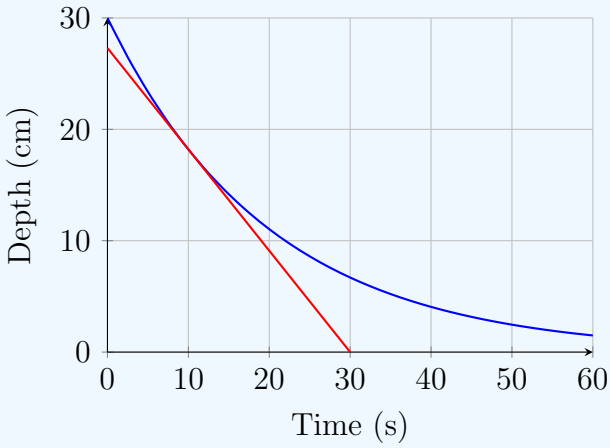
Look at the axes. If y is **Depth (cm)** and x is **Time (s)**, the gradient is **cm/s**. If y is **Volume** (m^3) and x is **Radius (m)**, the gradient is m^2.

→ Step 2: Direction Matters

- Graph going **UP** → Positive Rate (Filling, Heating).
- Graph going **DOWN** → Negative Gradient (Emptying, Cooling).
- **Warning:** If the question asks "Calculate the *rate of decrease*", usually give the **positive** number (magnitude). If it asks for "gradient", give the negative number.

Worked Example: Leakage (Rate of Decrease)

Problem: The graph shows the depth of liquid in a container. Estimate the rate of decrease of depth at $t = 10$ seconds.



Solution: 1. Draw tangent at $t = 10$. 2. Pick points: $(0, 27.3)$ and $(20, 9.1)$.

$$\text{Gradient} = \frac{9.1 - 27.3}{20 - 0} = \frac{-18.2}{20} = -0.91$$

3. Question asks for "rate of decrease".

Final Answer

Rate = 0.91 cm/s

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ZONE 4: The Explorer (Advanced Concepts)

Recognition Zone 4

Zone Name: The Explorer

- **Trigger:** Questions about "Average rate", "Instantaneous rate", "Zero gradient", or matching Average to Instantaneous.
- **Keywords:** "Turning points", "Parallel", "Where is the gradient zero?".
- **Goal:** Understand the *shape* and connections.

If you see these → You are in Zone 4.

→ Step 1: Zero Gradient

Gradient is **zero** at the peaks (maximums) and troughs (minimums). The tangent is a flat horizontal line.

→ Step 2: Average vs Instantaneous

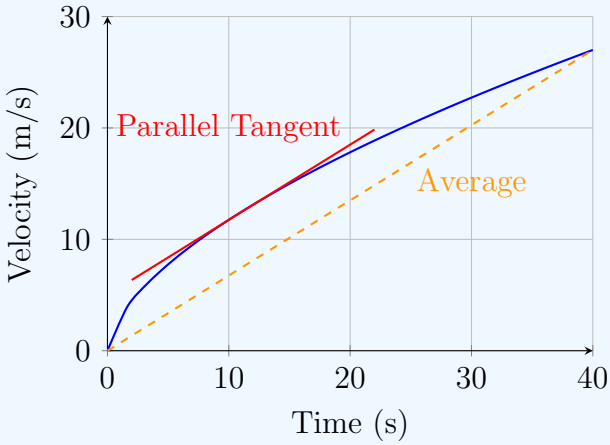
- **Instantaneous Rate:** Gradient of the **tangent** at a specific point.
- **Average Rate:** Gradient of the **chord** (secant) connecting two points (Start and End).

→ Step 3: The Parallel Match (Very Hard)

If asked: "Find the time T where instantaneous acceleration equals average acceleration", you must: 1. Draw a line connecting the start and end points (Average). 2. Place your ruler **parallel** to that line. 3. Slide the ruler until it touches the curve. That touch point is T.

Worked Example: Matching Average & Instantaneous

Problem: The graph shows velocity for 40 seconds. (a) Calculate average acceleration for the 40s. (b) Find t where instantaneous acceleration = average acceleration.



Solution: 1. **Average:** Connect $(0, 0)$ to $(40, 27)$.

$$m_{avg} = \frac{27 - 0}{40 - 0} = 0.675 \text{ m/s}^2$$

2. **Instantaneous:** Slide ruler parallel to the dashed line. It touches the blue curve at roughly $t = 12$.

Final Answer

$t \approx 12$ seconds

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EXAM SAFETY CHECKS

1. **Did you draw the line?** Never calculate without drawing the tangent first.
2. **Are units squared?** Acceleration is m/s^2. Speed is m/s.
3. **Is it negative?** If the line goes down left-to-right, your answer MUST be negative (unless asking for "decrease").
4. **Scale Check:** 1 large square usually $\neq$ 1 unit. Check if 1 square = 2, 5, or 10 units!

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7 Real-Life Graphs

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ZONE MASTERY GUIDE

REAL-LIFE GRAPHS (Kinematics & Conversion)

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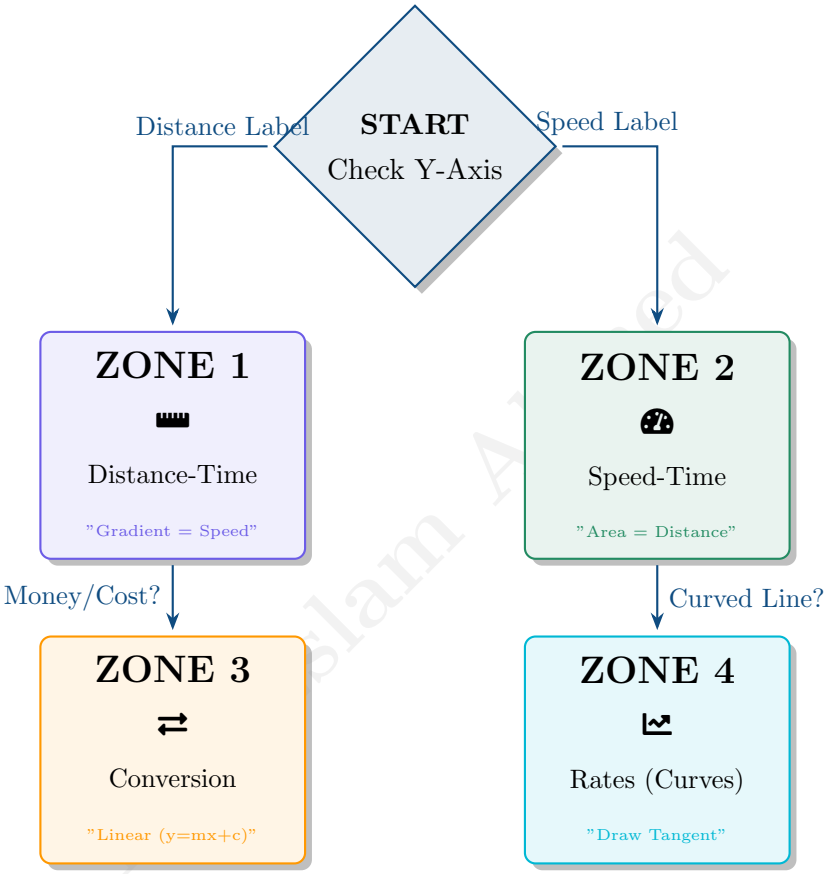
The 5-Second Rule: Match the Trigger → Pick the Zone.



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ZONE 1: The Travel Tracker (Distance-Time)

Recognition: Zone 1

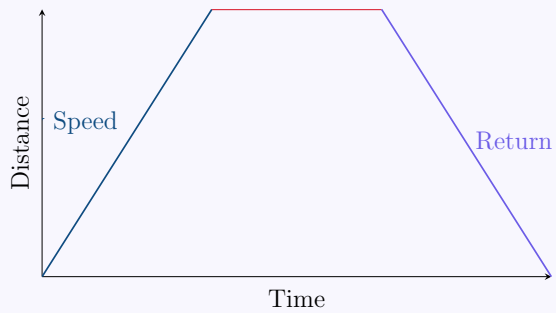
Zone Name: Distance-Time Graphs

- **Y-Axis:** Distance (m, km, miles).
- **X-Axis:** Time (seconds, minutes, hours).
- **Visual:** Straight lines sloping up, down, or flat.

If you see these → You are in Zone 1.

Strategic Breakdown

In this zone, the graph tells a story of a journey. You are tracking position over time.



- **Gradient (Slope) = Speed.** The steeper the line, the faster the object.
- **Horizontal Line = Rest.** The object has stopped.
- **Negative Gradient = Returning.** Moving back to the start.

Veteran Tip

The "Average Speed" Trap:
Never just add the speeds and divide by 2!
Formula: Average Speed $= \frac{\text{Total Distance}}{\text{Total Time}}$

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Worked Example (Adapted from PDF Q1)

Simon's Cycle Ride

Simon went for a cycle ride. He left home at 2 pm. The graph shows:

- 2 pm to 3 pm: Travels 12 km away.
- 3 pm to 4 pm: Horizontal line at 12 km.
- 4 pm to 5 pm: Travels from 12 km to 22 km.

(a) At 3 pm Simon stopped for a rest. How many minutes did he rest?
(b) How far was Simon from home at 5 pm?
(c) Calculate Simon's average speed for the first 3 hours (2 pm to 5 pm).

→ Step 1: Analyze the "Rest" Phase

Rest is represented by a horizontal line. The line is horizontal from 3 pm to 4 pm.

$$4 \text{ pm} - 3 \text{ pm} = 1 \text{ hour} = 60 \text{ minutes}$$

→ Step 2: Read the Graph for Distance

Go to 5 pm on the x-axis. Go up to the line, then across to the y-axis. The y-value is **22 km**.

→ Step 3: Calculate Average Speed

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$

Total Distance at 5 pm = 22 km. Total Time (2 pm to 5 pm) = 3 hours.

$$\text{Speed} = \frac{22}{3} = 7.33 \text{ km/h}$$

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ZONE 2: The Velocity Vault (Speed-Time)

Recognition: Zone 2

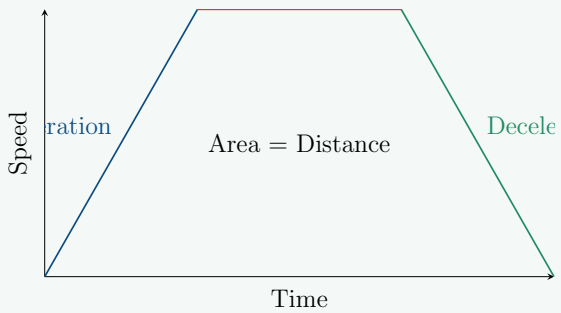
Zone Name: Speed-Time Graphs

- **Y-Axis:** Speed or Velocity (m/s, km/h).
- **X-Axis:** Time.
- **Visual:** Trapeziums, triangles, or rectangles.

If you see these → You are in Zone 2.

Strategic Breakdown

This graph shows *how fast* you are going, not where you are.



- **Gradient = Acceleration.** ($\frac{\text{Change in Speed}}{\text{Time}}$).
- **Horizontal Line = Constant Speed** (Acceleration is 0).
- **Area Under Graph = Distance Travelled.** This is the most common exam question type in this zone.

Common Traps

Do NOT confuse "Stop" with "Constant Speed"!
In Zone 1 (Distance-Time), a flat line means stopped.
In Zone 2 (Speed-Time), a flat line means you are still moving, just not getting faster! To stop in Zone 2, the line must go down to the x-axis ($Speed = 0$).

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Worked Example (Adapted from PDF Q13)

The Tram Journey

A tram travels from the library to the town hall.

- Accelerates uniformly from rest to 6 m/s over 10 seconds (implied).
- Travels at constant speed of 6 m/s.
- Decelerates from 6 m/s to rest between $t = 65$ and $t = 85$.

(a) Calculate the deceleration.
(b) Calculate the total distance travelled.

→ Step 1: Calculate Deceleration (Gradient)

Deceleration happens at the end (downward slope).

$$\text{Gradient} = \frac{\text{Rise}}{\text{Run}} = \frac{\text{Change in Speed}}{\text{Time}}$$

Speed goes from 6 to 0. Time goes from 65 to 85 (20 seconds).

$$\text{Deceleration} = \frac{6}{20} = 0.3 \text{ m/s}^2$$

→ Step 2: Calculate Distance (Area)

Split the shape into simple polygons or use the Trapezium Rule.

Method A: Shapes

1. Triangle 1 (Start): (Implied base 10s). Area $= \frac{1}{2} \times 10 \times 6 = 30$ m.
2. Rectangle (Middle): Area = width × height.
3. Triangle 2 (End): Area $= \frac{1}{2} \times 20 \times 6 = 60$ m.

Method B: Trapezium Formula (Best for entire graph)

$$\text{Area} = \frac{1}{2}(a + b)h$$

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ZONE 3: The Currency Converter (Linear Models)

Recognition: Zone 3

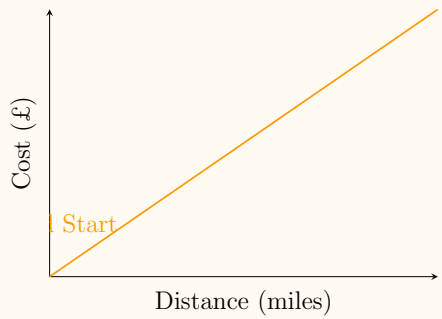
Zone Name: Conversion & Cost Graphs

- **Y-Axis & X-Axis:** Different units (e.g., \$ vs £, Distance vs Cost).
- **Visual:** A straight line that might *not* start at zero.
- **Keywords:** "Fixed charge", "Call-out fee", "Rate per mile".

If you see these → You are in Zone 3.

Strategic Breakdown

These graphs represent linear equations of the form $y = mx + c$.



- **Y-Intercept (c):** The **Fixed Charge** or initial value. (Cost before you even move).
- **Gradient (m):** The **Rate**. (e.g., Cost *per* mile, Exchange rate).
- **Direct Proportion:** If the line passes through $(0, 0)$, y is directly proportional to x.

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Worked Example (Adapted from PDF Q5)

The Delivery Van

Bill delivers parcels. There is a fixed charge plus a cost per mile. The graph shows the cost (C) against distance (d).

- The line starts at £10 on the y-axis.
- At 10 miles, the cost is £20.

(a) How much is the fixed charge?
(b) Calculate the cost per mile.

→ Step 1: Identify the Intercept

Look at where the line crosses the vertical axis (Distance = 0). The graph starts at £10.
Fixed Charge = £10.

→ Step 2: Calculate the Gradient

Pick two clear points on the line. Point 1: $(0, 10)$ Point 2: $(10, 20)$

$$\text{Gradient} = \frac{\text{Change in Cost}}{\text{Change in Distance}}$$

$$\text{Gradient} = \frac{20 - 10}{10 - 0} = \frac{10}{10} = 1$$

Answer: £1.00 per mile.

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ZONE 4: The Curve Commander (Rates of Change)

Recognition: Zone 4

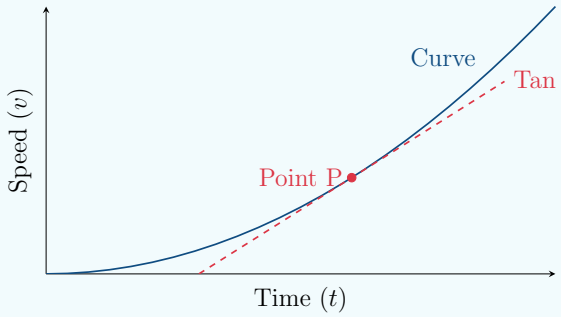
Zone Name: Curved Graphs & Tangents

- **Visual:** The line is curved (non-linear).
- **Question asks:** "Calculate the instantaneous speed" or "acceleration at $t = 5$".
- **Context:** Filling containers with water, or variable acceleration.

If you see these → You are in Zone 4.

Strategic Breakdown

When the line curves, the speed (or rate) is changing every second. You cannot just do distance divided by time.



- **Average Rate:** Draw a chord (straight line) connecting start and end points. Find its gradient.
- **Instantaneous Rate:** Draw a **Tangent** (a straight line just touching the curve) at the specific time. Find the gradient of the tangent.

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Veteran Tip

The Water Container Trick (PDF Q17)

Graph gradient = How fast the water level rises.

- **Wide Container:** Water rises slowly → Shallow Gradient.
- **Narrow Container:** Water rises quickly → Steep Gradient.
- **Vase (Wide to Narrow):** Slow then Fast → Curve gets steeper.

Worked Example (Adapted from PDF Q11 - Hard)

Instantaneous Acceleration

The graph shows the speed of a car. It is a curve. **Use the graph to estimate the acceleration at $t = 7$ seconds.**

Step 1: Draw a Tangent

Place your ruler at $t = 7$ on the curve. Adjust the angle so the ruler touches the curve at that one point but doesn't cut through it. Draw a long straight line.

Step 2: Pick Points on the Tangent

Do NOT use points on the curve. Use points on your *straight red line*. Let's say your tangent passes through $(4, 10)$ and $(8, 30)$.

Step 3: Calculate Gradient

$$\text{Acceleration} = \frac{30 - 10}{8 - 4} = \frac{20}{4} = 5 \text{ m/s}^2$$

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CHALLENGE MODE: Mixed Concepts

Common Traps

The "Units" Trap in Areas

When calculating distance from a speed-time graph, check the axes!

- x-axis in **minutes**, y-axis in **km/h**.
- You MUST convert minutes to hours before multiplying.
- Example: Speed 60 km/h for 15 minutes.
- Wrong: $60 \times 15 = 900$ km.
- Right: 15 mins = 0.25 hours. $60 \times 0.25 = 15$ km.

Hard: Variable Acceleration (PDF Q12)

A vehicle travels for 50 seconds.

- 0 to 40s: Constant speed of k m/s.
- 40 to 50s: Decelerates to rest.
- Total Distance = 1.35 km.

Work out the value of k.

→ Step 1: Convert Units

Distance is given in km, speed in m/s. Convert distance to meters.

$$1.35 \text{ km} = 1350 \text{ meters}$$

→ Step 2: Set up Area Formula

The graph is a Trapezium (or Rectangle + Triangle). Area = Distance.

Using Trapezium Rule:

$$\text{Area} = \frac{1}{2}(a+b)h$$

$$a = 40 \text{ (top flat line)}$$

$$b = 50 \text{ (bottom total time)}$$

$$h = k \text{ (the unknown speed)}$$

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→ **Step 3: Solve Equation**

$$1350 = \frac{1}{2}(40 + 50) \times k$$

$$1350 = 0.5 \times 90 \times k$$

$$1350 = 45k$$

$$k = \frac{1350}{45}$$

$$k = 30 \text{ m/s}$$

Final Answer

k = 30

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8 Graphing Inequalities

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GRAPHING INEQUALITIES

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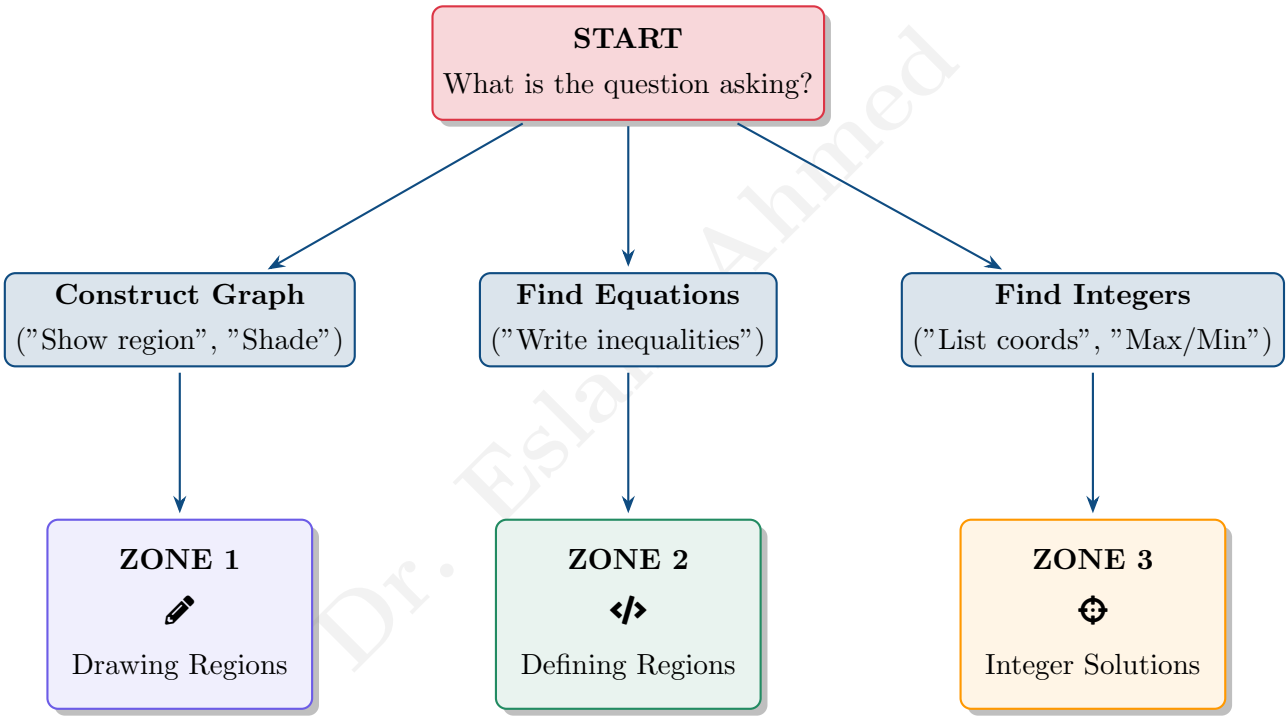
The 5-Second Rule: Match the Trigger → Pick the Zone.

Zone Navigation Map



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ZONE 1: The Architect (Drawing Regions)

Recognition: Zone 1

Zone Name: Drawing Regions

- **Keywords:** "Show by shading", "Shade the unwanted region", "Draw the region".
- **Visual:** You are given a blank grid or empty axes and algebraic inequalities (e.g., $y \leq 2x + 1$).
- **Goal:** Turn Algebra $\rightarrow$ Geometry.

If you see these $\rightarrow$ You are in Zone 1.

The "Border & Test" Strategy

Step 1: Treat Inequality as Equation

Ignore the $<, >, \leq, \geq$ symbols initially. Replace them with $=$ to find the boundary line.

- $x \leq 4 \rightarrow$ Draw line $x = 4$ (Vertical).
- $y > 3 \rightarrow$ Draw line $y = 3$ (Horizontal).
- $y \geq 2x + 1 \rightarrow$ Draw line $y = 2x + 1$ (Diagonal).

Step 2: Solid or Dashed?

Look at the original symbol:

- **Solid Line:** For $\leq$ or $\geq$ (The boundary IS included).
- **Dashed Line:** For $<$ or $>$ (The boundary IS NOT included).

Step 3: The "Test Point" Technique

Pick a point **NOT** on the line (usually $(0, 0)$ if the line doesn't pass through origin).

- Substitute into the inequality.
- **TRUE?** The region containing the point is the **WANTED** region.
- **FALSE?** The region containing the point is the **UNWANTED** region.

Step 4: Shade Correctly (Crucial!)

READ THE QUESTION: Does it say "Shade the region" (wanted) or "Shade the unwanted region"?

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- IGCSE Standard: Usually asks to "Shade the **unwanted** region" so the clear white space is the answer R.
- Modern Papers: Sometimes ask to "Shade the region R". **Always check!**

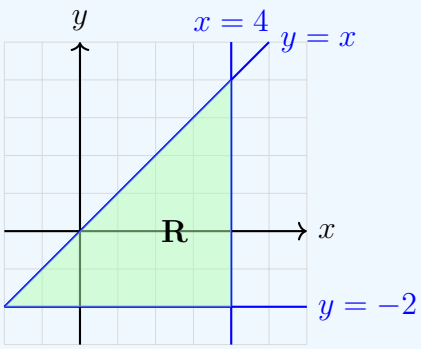
Worked Example (Easy): Standard Shading

Problem: Show, by shading on the grid, the region that satisfies:

$$x \leq 4 \quad \text{and} \quad y \geq -2 \quad \text{and} \quad y \leq x$$

Solution:

1. **Line 1 ($x = 4$):** Vertical line at $x = 4$. Solid ($\leq$). Wanted is Left.
2. **Line 2 ($y = -2$):** Horizontal line at $y = -2$. Solid ($\geq$). Wanted is Above.
3. **Line 3 ($y = x$):** Diagonal line through $(0, 0), (2, 2)$. Solid ($\leq$). Test $(2, 0)$: $0 \leq 2$ is True. So wanted is below.



Veteran Tip

If you have to shade the **unwanted** regions: Draw little "hairs" or distinct shading lines coming off the line on the "bad" side immediately after drawing it. This keeps the "good" region clear.

Common Traps

- **Dashed vs Solid:** Losing 1 mark by drawing a solid line for $y < 2x$. Always check the "equals" bar!
- **Shading the Wrong Side:** Don't guess. Test $(0, 0)$. If $0 > 3$ is FALSE, shade the side with $(0, 0)$ if shading unwanted.
- **x vs y:** Drawing $y = 3$ as a vertical line. Remember: $y =$ cuts the y-axis (horizontal).

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ZONE 2: The Interpreter (Defining Regions)

Recognition: Zone 2

Zone Name: Defining Regions

- **Keywords:** "Write down the three inequalities", "Find the inequalities that define region R".
- **Visual:** You see a shaded shape (triangle, quadrilateral) bounded by lines.
- **Goal:** Geometry $\rightarrow$ Algebra.

If you see these $\rightarrow$ You are in Zone 2.

The "Equation Extraction" Strategy

Step 1: Identify the Lines

Extend the boundary lines of the shape visually to the axes.

- Vertical line crossing x-axis at k? $\rightarrow x = k$.
- Horizontal line crossing y-axis at k? $\rightarrow y = k$.
- Diagonal? Find gradient $m = \frac{\text{rise}}{\text{run}}$ and intercept c. Equation: $y = mx + c$.

Step 2: Determine the Direction

Pick a point clearly **INSIDE** the shaded region (e.g., region R). Let's say $(2, 1)$. Substitute into the equation sides:

- If line is $y = x$, compare y coord (1) with x coord (2). $1 < 2$, so inequality is $y < x$.
- If line is $x = 4$, look at the region. Is it left or right? Left $\rightarrow x < 4$.

Step 3: Set the Boundary Type

- Solid Line $\rightarrow$ Add "or equal to" $(\leq, \geq)$.
- Dotted Line $\rightarrow$ Strict inequality $(<, >)$.

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Worked Example (Medium): Triangle Region

Problem: A region R is bounded by $x = 1$, $y = -3$ and a diagonal line passing through $(1, -3)$ and $(0, -1)$. Write the inequalities.

Solution:

1. **Line 1 ($x = 1$):** The region is to the left? No, usually bounded means "enclosed". Let's assume the triangle is formed by lines.
2. **Diagonal Line:** Passes through $(0, -1)$ so $c = -1$. Gradient $m = \frac{-1-(-3)}{0-1} = \frac{2}{-1} = -2$. Equation: $y = -2x - 1$.
3. **Finding Signs:** Pick a point in the center, say $(0, -2)$.
4. Check $x = 1$: Point has $x = 0$. $0 < 1$. So $x \leq 1$.
5. Check $y = -3$: Point has $y = -2$. $-2 > -3$. So $y \geq -3$.
6. Check $y = -2x - 1$: Point $(0, -2)$. LHS $= -2$. RHS $= -1$. $-2 < -1$. So $y \leq -2x - 1$.

Final Answer

$$x \leq 1, \quad y \geq -3, \quad y \leq -2x - 1$$

Common Traps

- **Gradient Sign Errors:** A line going *down* from left to right has a negative gradient.
- **Confusing x and y:** For vertical lines, the equation is $x = ...$, not $y = ...$
- **Assuming $\leq$:** If the diagram clearly shows a dotted line, you MUST use $<$ or $>$.

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ZONE 3: The Hunter (Integer Solutions)

Recognition: Zone 3

Zone Name: Integer Solutions

- **Keywords:** "Find coordinates of integers", "List possible values of x", "Find point inside R".
- **Visual:** Usually combined with a grid or a set of inequalities.
- **Goal:** Find specific (x, y) pairs where $x, y \in \mathbb{Z}$.

If you see these → You are in Zone 3.

The "Search & Check" Strategy

→ Step 1: Define the Search Area
Identify the rough geometric bounds (e.g., x is between 1 and 4, y is between 0 and 5).

→ Step 2: List Candidate Integers
Write down all integer x values allowed. For each x, check the allowed range of y.

→ Step 3: Boundary Check (The Trap)
If the inequality is strict ($<$ or $>$), points ON the line are **NOT** solutions. If the inequality is inclusive ($\leq$ or $\geq$), points ON the line **ARE** solutions.

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Worked Example (Hard): Finding Points

Problem: Find the point (x, y) with integer coordinates inside the region R defined by:

$$3x + 5y = 35 \quad \text{(Equation of boundary)}$$

$$y > 1, \quad x \geq 2, \quad \text{and is below the line } 3x + 5y = 35$$

Solution: This is often a trial and error within limits.

1. **Constraint 1:** $x \geq 2$. Candidates: $x = 2, 3, 4, 5...$
2. **Constraint 2:** $y > 1$. Candidates: $y = 2, 3, 4...$
3. **Constraint 3:** Below $3x + 5y = 35$, so $3x + 5y < 35$.
4. **Trial 1 ($x = 2$):** $3(2) + 5y < 35 \to 6 + 5y < 35 \to 5y < 29 \to y < 5.8$. Integers $y > 1$: $y \in \{2, 3, 4, 5\}$.
5. **Trial 2 ($x = 3$):** $3(3) + 5y < 35 \to 9 + 5y < 35 \to 5y < 26 \to y < 5.2$. Integers $y > 1$: $y \in \{2, 3, 4, 5\}$.
6. **Specific Question Context:** Sometimes there's only ONE valid point or a specific optimization required (e.g., "max value of x+y").

If the question asks for a point ON the line $3x + 5y = 35$ inside constraints: Check $x = 5$: $15 + 5y = 35 \to 5y = 20 \to y = 4$. Point $(5, 4)$. Check $x \geq 2$: Yes $(5 \geq 2)$. Check $y > 1$: Yes $(4 > 1)$.

Final Answer

Point: $(5, 4)$

Veteran Tip

If asked for "Integer points satisfying...": 1. Always draw a table: Columns for x and allowed y. 2. Watch out for 0. It is an integer! 3. Watch out for negative integers if the graph allows.

Common Traps

- **Boundary Confusion:** Including a point on a dotted line. E.g., if $x < 3$, then $x = 3$ is NOT a solution.
- **Coordinate Swaps:** Listing (y, x) instead of (x, y).

9 Transformations of Graphs

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Graph Transformations

Topic: $y = f(x)$ adjustments

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November 27, 2025

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The 5-Second Rule:
Match the **Trigger** (Symbol) → Pick the **Zone** (Strategy).

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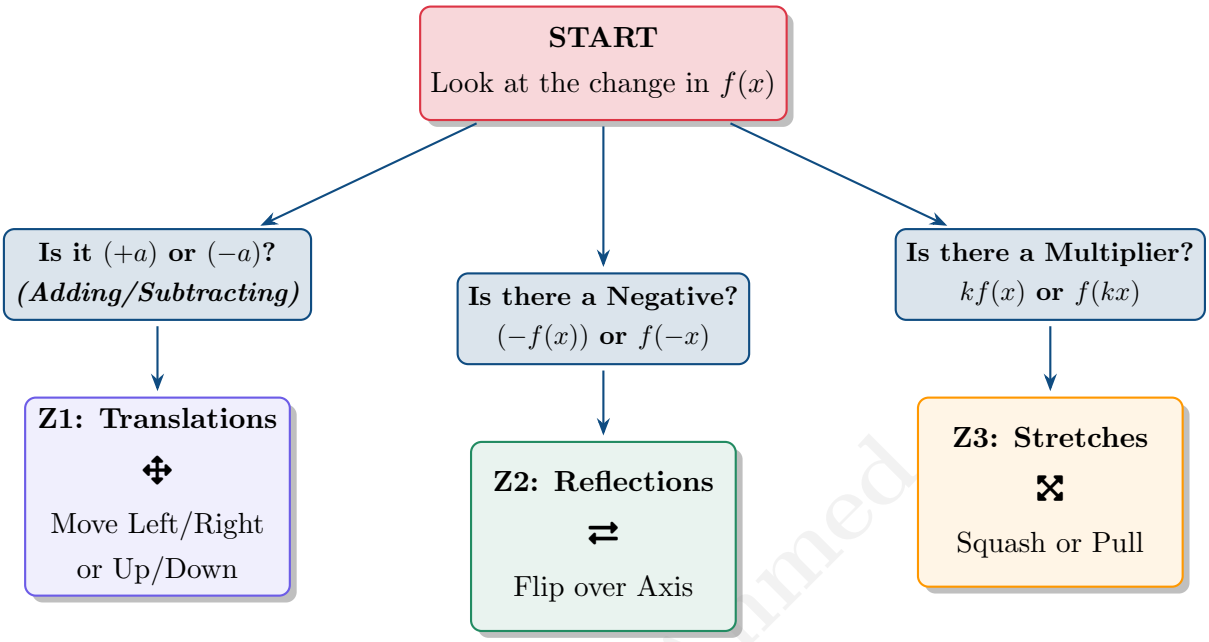
Zone Navigation Map

Use this map to decide which strategy to apply.



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ZONE 1: Translations (The Slide)

Recognition: Zone 1

Zone Name: Translations

Look for **Addition** or **Subtraction**.

- $y = f(x) + a$
- $y = f(x + a)$

If you see this → You are in Zone 1.

Step 1: Check: Inside or Outside?

- **Outside** bracket: $f(x) + a \rightarrow$ Affects y (Vertical).
- **Inside** bracket: $f(x + a) \rightarrow$ Affects x (Horizontal).

Step 2: Apply the "Truth vs Liar" Rule

- **Outside is Truthful**: $+a$ moves UP, $-a$ moves DOWN.
- **Inside is a Liar**: $+a$ moves LEFT, $-a$ moves RIGHT.

Veteran Tip

If you see $y = f(x - 3)$, your brain says "Left" because of the minus. **STOP!** Inside is a Liar. It moves **RIGHT** by 3.

Worked Example

Problem: The curve $y = f(x)$ has a minimum at $(2, -5)$. Find the new minimum of $y = f(x + 4)$.

Solution:

1. The $+4$ is **Inside**.
2. Inside = Horizontal Liar.
3. $+4$ means move **Left** by 4 (subtract 4 from x).
4. New $x = 2 - 4 = -2$. y stays -5.

Final Answer

$(-2, -5)$

ZONE 2: Reflections (The Flip)

Recognition: Zone 2

Zone Name: Reflections

Look for a **Negative Sign** multiplying the function.

- $y = -f(x)$
- $y = f(-x)$

If you see this → You are in Zone 2.

Step 1: Locate the Negative

Is it **Outside** the function or **Inside** next to the x?

Step 2: Determine the Axis

- **Outside** $(-f(x))$: Affects y. Flip vertically over the **X-axis**.
- **Inside** $(f(-x))$: Affects x. Flip horizontally over the **Y-axis**.

Quick Check

Curve passes through $(3, 10)$.

A) $y = -f(x) \to$ Change sign of $y \to (3, -10)$.

B) $y = f(-x) \to$ Change sign of $x \to (-3, 10)$.

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ZONE 3: Stretches (Squash & Pull)

Recognition: Zone 3

Zone Name: Stretches

Look for a **Multiplier** (Scale Factor).

- $y = af(x)$
- $y = f(ax)$

If you see this → You are in Zone 3.

Step 1: Vertical vs Horizontal

- **Outside** ($2f(x)$): Vertical Stretch. Multiply y by 2.
- **Inside** ($f(2x)$): Horizontal Stretch. Divide x by 2 (Multiply by $\frac{1}{2}$).

Veteran Tip

$f(2x)$ feels like it should get bigger, but it gets **Squashed**! Think of it as "speeding up" the graph, so everything happens twice as fast (half the space).

Exam Style Question

The point $(4, 8)$ lies on $y = f(x)$. Find the coordinate on $y = f(2x)$.

Solution:

1. The 2 is inside. Inside is a Liar.
2. Operation: Divide x by 2. Keep y same.
3. $x_{new} = 4 \div 2 = 2$.

Final Answer

$(2, 8)$

10 Differentiation

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code Code

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DIFFERENTIATION: IGCSE MASTER CLASS

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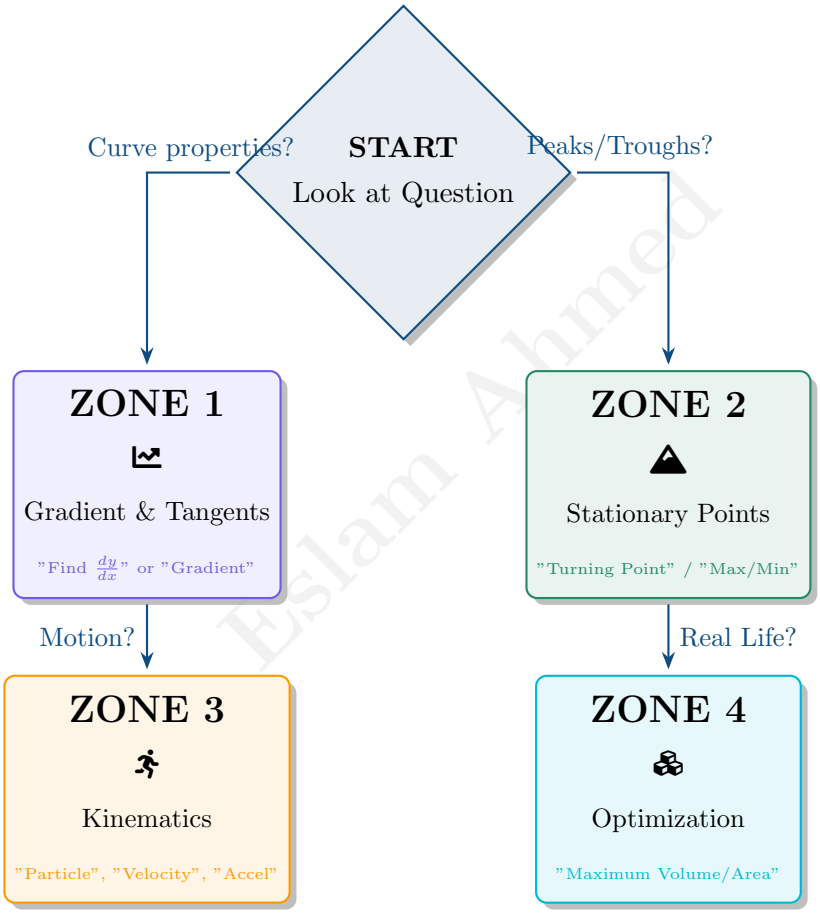
The 5-Second Rule: Match the Trigger → Pick the Zone.



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ZONE 1: The Gradient Machine

1 ZONE 1: The Gradient Machine

Differentiation converts a curve equation (y) into a gradient finder ($\frac{dy}{dx}$).

Recognition: Zone 1

Zone Name: Basic Differentiation **Trigger 1:** "Find $\frac{dy}{dx}$".

Trigger 2: "Find the gradient of the curve at...".

Trigger 3: "Find the coordinates where the gradient is...". **If you see these $\rightarrow \rightarrow$ You are in Zone 1.**

1.1 Strategy: The Power Rule

→ Step 1: Prepare the Terms Ensure every term is in the form ax^n.

- $\frac{1}{x} \rightarrow x^{-1}$
- $\sqrt{x} \rightarrow x^{0.5}$

→ Step 2: Differentiate Multiply by the power, reduce the power by 1.

$$y = ax^n \implies \frac{dy}{dx} = anx^{n-1}$$

Note: Constants (numbers without x) disappear. $5x$ becomes 5. **→ Step 3: Substitute (If asked)** If finding a gradient at a specific point, replace x with the coordinate.

1.2 Worked Examples

Level 1: Basic Derivation (PDF Q3)

Problem: $y = 2x^3 - 6x^2 + 3x - 4$. Find $\frac{dy}{dx}$.
Solution:

1. Term 1: $2x^3 \to 3 \times 2x^{3-1} = 6x^2$.
2. Term 2: $-6x^2 \to 2 \times -6x^{2-1} = -12x$.
3. Term 3: $3x \to 3$.
4. Term 4: $-4 \to 0$ (Constant disappears).

Final Answer

$$\frac{dy}{dx} = 6x^2 - 12x + 3$$

Level 2: Working Backwards (PDF Q4)

Problem: Curve $y = 2x^2 - 6x - 11$. Find the coordinates of the point where the gradient is 2.
Solution:

1. Find Gradient Function: $\frac{dy}{dx} = 4x - 6$.
2. Trigger: "Gradient is 2". Set $\frac{dy}{dx} = 2$.

$$4x - 6 = 2 \implies 4x = 8 \implies x = 2$$

3. Find Coordinates: We need y. Put $x = 2$ back into original equation.

$$y = 2(2)^2 - 6(2) - 11 = 8 - 12 - 11 = -15$$

Final Answer

$(2, -15)$

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Common Traps

The Constant Trap: Students often leave the constant.

- Wrong: $y = x^2 + 5 \to \frac{dy}{dx} = 2x + 5$
- Right: $y = x^2 + 5 \to \frac{dy}{dx} = 2x$

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ZONE 2: Stationary Points (Max/Min)

2 ZONE 2: Stationary Points

At the top of a hill (max) or bottom of a valley (min), the ground is flat. The gradient is zero.

Recognition: Zone 2

Zone Name: Turning Points **Trigger 1:** "Find the stationary points".

Trigger 2: "Find the turning points".

Trigger 3: "Find the maximum/minimum value". **If you see these → →**

You are in Zone 2.

2.1 Strategy: The Zero Gradient Protocol

→ **Step 1: Differentiate** Find $\frac{dy}{dx}$. → **Step 2: Equate to Zero** Set $\frac{dy}{dx} = 0$. This is the "Stationary Condition". → **Step 3: Solve for x** Solve the resulting equation (usually linear or quadratic). → **Step 4: Find y** Substitute x back into the **original** $y = \ldots$ equation to get coordinates.

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2.2 Worked Examples

Level 1: Finding Points (PDF Q12)

Problem: $y = x^3 - 6x^2 - 15x$. Find the coordinates of the two stationary points.
Solution:

1. Differentiate: $\frac{dy}{dx} = 3x^2 - 12x - 15$.
2. Set to 0: $3x^2 - 12x - 15 = 0$.
3. Simplify: Divide by 3 $\rightarrow x^2 - 4x - 5 = 0$.
4. Factorise: $(x-5)(x+1) = 0$.
5. Roots: $x = 5$ and $x = -1$.
6. Find y coordinates:
   - At $x = 5$: $y = 5^3 - 6(5)^2 - 15(5) = 125 - 150 - 75 = -100$.
   - At $x = -1$: $y = (-1)^3 - 6(-1)^2 - 15(-1) = -1 - 6 + 15 = 8$.

Final Answer

$(5, -100)$ and $(-1, 8)$

Veteran Tip

Classification Check: How do you know if it's Max or Min?

- **Method 1 (Shape):** Positive Cubic ($+x^3$) goes Up-Down-Up. So first point (left) is Max, second (right) is Min.
- **Method 2 (Second Derivative):** Find $\frac{d^2y}{dx^2}$. If positive $\rightarrow$ Minimum (Happy face). If negative $\rightarrow$ Maximum (Sad face).

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ZONE 3: Kinematics (The Motion Trio)

3 ZONE 3: Kinematics

Motion is defined by 3 variables: Displacement (s), Velocity (v), and Acceleration (a). Differentiation links them.

Recognition: Zone 3

Zone Name: Movement **Trigger 1:** "Particle P moves along a straight line".

Trigger 2: Equations with s and t (e.g., $s = 6t^3 \dots$).

Trigger 3: "Instantaneously at rest" (Means $v = 0$). **If you see these → →**

You are in Zone 3.

3.1 Strategy: The Ladder

Displacement (s)

↓ Diff ($\frac{ds}{dt}$)

Velocity (v)

↓ Diff ($\frac{dv}{dt}$)

Acceleration (a)

Dr. Es

Dr. Eslam Ahmed

3.2 Worked Examples

Level 1: Finding a (PDF Q6)

Problem: $s = 6t^3 - 12t^2 + 7t$. Find the time when acceleration is 3 m/s^2.
Solution:

1. Find Velocity ($v = \frac{ds}{dt}$):
$$v = 18t^2 - 24t + 7$$
2. Find Acceleration ($a = \frac{dv}{dt}$):
$$a = 36t - 24$$
3. Trigger: "Acceleration is 3".
$$36t - 24 = 3$$
$$36t = 27 \implies t = \frac{27}{36} = \frac{3}{4}$$

Final Answer

$t = 0.75$ seconds

Level 2: Instantaneous Rest (PDF Hard Q9)

Problem: $s = t^3 + 5t^2 - 8t + 10$. Find displacement when P is at rest.
Solution:

1. "At rest" means $v = 0$.
2. Find v: $v = 3t^2 + 10t - 8$.
3. Solve $3t^2 + 10t - 8 = 0$.
$$(3t - 2)(t + 4) = 0 \implies t = \frac{2}{3} \text{ or } t = -4$$
4. Discard negative time. $t = \frac{2}{3}$.
5. Substitute $t = \frac{2}{3}$ back into s equation.

ZONE 4: Optimization (Real World Max/Min)

4 ZONE 4: Optimization

Using differentiation to find the "best" amount (Maximum Area, Minimum Cost, Maximum Volume).

Recognition: Zone 4

Zone Name: Modeling **Trigger 1:** Shapes (Rectangles, Cuboids) with variable sides (x).
Trigger 2: "Show that the Volume $V = \ldots$".
Trigger 3: "Find the value of x for which V is a maximum". **If you see these $\rightarrow \rightarrow$ You are in Zone 4.**

4.1 Strategy: The 3-Step Opti-Flow

→ Step 1: Construct the Formula (The "Show That") Usually involves substitution.

- Example: Perimeter $= 40$. $2x + 2y = 40 \implies y = 20 - x$.
- Area $= xy = x(20 - x) = 20x - x^2$.

→ Step 2: Differentiate Treat variables like V or A as y. Find $\frac{dV}{dx}$ or $\frac{dA}{dx}$. **→ Step 3: Maximize/Minimize** Set the derivative to 0 and solve.

4.2 Worked Examples

Level 1: Garden Fence (PDF Q7)

Problem: A homeowner has 40m of fence for 3 sides of a rectangle (one side is a wall). Width w, Length l.

- (a) Show $A = 40w - 2w^2$.
- (b) Find w for max area.

Solution:

1. Constraint: Fence used for 2 widths + 1 length. $2w+l = 40 \implies l = 40-2w$.
2. Area: $A = w \times l = w(40 - 2w) = 40w - 2w^2$.
3. Maximize: Find $\frac{dA}{dw}$.
$$\frac{dA}{dw} = 40 - 4w$$
4. Set to 0: $40 - 4w = 0 \implies 4w = 40 \implies w = 10$.

Final Answer

$w = 10$ m

Level 2: Cuboid Volume (PDF Q8)

Problem: $V = 15 + 16x - x^2 - 2x^3$. Find x for max volume.
Solution:

1. Differentiate: $\frac{dV}{dx} = 16 - 2x - 6x^2$.
2. Set to 0: $16 - 2x - 6x^2 = 0$.
3. Simplify: $3x^2 + x - 8 = 0$ (Divide by -2 and rearrange).
4. Solve (Quadratic Formula):
$$x = \frac{-1 \pm \sqrt{1^2 - 4(3)(-8)}}{6} = \frac{-1 \pm \sqrt{97}}{6}$$
5. Pick positive x: $x \approx 1.47$.

Final Answer

$x = 1.47$

Dr. Eslam Ahmed

Conclusion

Differentiation is just a tool to study change.

1. **Geometry:** It finds Gradients.
2. **Graphs:** It finds Turning Points.
3. **Physics:** It links Displacement $\rightarrow$ Velocity $\rightarrow$ Acceleration.
4. **Business/Design:** It finds the Maximum or Minimum possible result.

Dr. Eslam Ahmed

Dr. Eslam Ahmed